Grade 7 General Science Solved Exercises - Balochistan Textbook Board

Complete Book – All 12 Units

Balochistan Textbook Board (BTBB)

Unit 1: Plant Systems

Exercise (p.21): Multiple Choice Questions — Encircle the correct option.

(i) In root system, surface area for absorption of water is increased by:

- Root cuticle
- ✓ Root hairs
- Root cap
- Phloem in root

(ii) One of the functions of cortex in roots is to:

- Prepare food
- ✓ Store food
- Transport food
- Digest food

(iii) Living organisms get energy from food. Where do plants get food from?

- Use air as food
- Use water as food
- Use soil as food
- ✓ Prepare their own food

(iv) Vascular bundles consist of:

- Epidermis and endodermis

Upper epidermis and lower epidermis

✓ Xylem and phloem

Primary roots and secondary roots

(v) The colour of chlorophyll is:

Red

Blue

Yellow

✓ Green

(vi) The element needed by plants for making chlorophyll is:

Iron

✓ Magnesium

Carbon

Sulphur

(vii) Which of the following is needed by plants for making proteins?

Oxygen

Carbon dioxide

✓ Nitrogen

Phosphorous

(viii) The process of releasing energy is known as:

✓ Respiration

Photosynthesis

Growth

Development

(ix) Food prepared in leaves is transported to other parts of the plant through:

- Xylem
- ✓ Phloem
- Epidermis
- Mesophyll

(x) Besides oxygen, the main product of photosynthesis is:

- ✓ Glucose
- Proteins
- Carbon dioxide
- Water

Exercise (p.22): 1.2 Write short answers.

1. Write names of the products of respiration.

The products of respiration are carbon dioxide, water and energy (ATP). The word equation is: Food (Glucose) + Oxygen → Carbon dioxide + Water + Energy (ATP).

2. Write names of the reactants of respiration.

The reactants of respiration are food (glucose) and oxygen. Oxygen reacts with food to release energy.

3. Define photosynthesis.

Photosynthesis is a process during which plants prepare their own food using water and carbon dioxide in the presence of sunlight and chlorophyll. Plants take in carbon dioxide from the air, water from the soil and sunlight from the environment to produce glucose

(sugar) and oxygen.

4. What are different parts of root system in plants?

The root system consists of the primary root (tap root), which originates from the radicle of the embryo and grows downwards in the soil; secondary roots, the smaller roots that grow from the primary root; and tertiary roots, the further branches of secondary roots. Plants like grasses and strawberry have threadlike fibrous roots spread in the soil but not growing deep. The root tip has the root meristem (rapidly dividing cells) covered and protected by the root cap, and root hairs — tiny hair-like outgrowths of the epidermis that increase the surface area for absorption of water and nutrients. The regions of the root are the region of meristematic activity, the region of elongation and the region of maturation.

5. Enlist the functions of shoot system in plants.

The shoot system is made of the above-ground parts of a plant: stem, leaves, flowers and fruit. Its functions are: to conduct water from roots to leaves for use in making food; to provide support to leaves and branches (the stem supports all other parts of the shoot system, and the internal vascular bundles connect the roots to the leaves by passing through the stem); and to transport food to every part of the plant body. Leaves are the main site of photosynthesis, gaseous exchange and removal of extra water; the stem also bears nodes (where a leaf is attached) and internodes (the part between two nodes).

Exercise (p.22): 1.3 Differentiate between the following.

1. Epidermis and endodermis

Epidermis is the outermost layer of the root or stem, made of thin-walled cells; in the root it bears unicellular outgrowths (root hairs) and protects the inner cells while playing a key role in absorption of water and nutrients from the soil, and in the stem it is coated on the outer side with a wax-like cuticle that prevents loss of water and protects against mechanical injuries. Endodermis is the innermost layer of the cortex, consisting of barrel-shaped cells; it regulates the movement of water and hormones into and out of the vascular bundles.

2. Transpiration and respiration

Transpiration is the process by which plants lose water vapour, mostly from their leaves, into the air — it is like a plant's way of sweating; when stomata open to let in carbon dioxide for photosynthesis, water vapour escapes from the moist surfaces inside the leaf. Respiration is the process during which oxygen reacts with food to release energy (ATP); it takes place in the mitochondria of every cell of animals and plants and is vital for producing energy from food.

3. Palisade mesophyll and spongy mesophyll

Palisade mesophyll lies just beneath the upper epidermis and consists of regularly arranged elongated cells provided with a large number of chloroplasts to facilitate photosynthesis. Spongy mesophyll is located near the lower epidermis and consists of irregularly shaped cells having air spaces among them to facilitate the exchange of gases.

4. Cohesion and adhesion

Cohesion is the force by which water molecules attract each other (intermolecular force); it helps water molecules move upwards in a continuous column from the roots all the way to the leaves. Adhesion is the force by which water sticks to the walls of the xylem tubes; liquids in narrow tubes (for example water in xylem) begin to move upwards due to adhesive force — this is known as the principle of transpiration.

5. Xylem and phloem tissues

Xylem is the tissue that conducts water and other nutrients from roots to leaves through the stem; its tubes are usually found towards the centre of the stem, have no end walls between cells (perforations), thick walls stiffened with lignin, and one-way flow of water and minerals. Phloem is the tissue that transports food prepared in leaves to other parts of the plant; its tubes are located closer to the outer edge of the stem and carry a two-way flow of water and food.

Exercise (p.22): 1.4 Descriptive Questions.

1. Illustrate the internal structures of root and stem and point out the differences between them.

Internal structure of root: epidermis (outermost layer with root hairs), cortex (many layers of thin-walled round cells inside the epidermis; water and nutrients absorbed by root hairs are transported to vascular bundles through the cortex, and reserve food is stored in it), endodermis (innermost layer of the cortex,

barrel-shaped cells regulating movement of water and hormones into and out of the vascular bundles), and vascular bundles (xylem and phloem), with the pericycle between the endodermis and the vascular bundles. Internal structure of stem: epidermis (external layer coated with cuticle, preventing water loss and mechanical injury), cortex (provides support to the stem and stores food), vascular bundles (xylem and phloem, tubular structures transporting water, minerals and food throughout the plant), cambium (a layer of tissue that produces new vascular tissues and allows the stem to grow in diameter), and pith (undifferentiated parenchyma cells in the centre of the stem, functioning in storage of nutrients and support; mainly present in early growth and often replaced by woody xylem cells in older branches and stems). Differences: the root has root hairs, an endodermis and a pericycle and no pith or cambium; the stem has a cuticle, a cambium layer for growth in diameter and a central pith; in the stem cross-section the xylem tubes are usually towards the centre while the phloem tubes are closer to the outer edge.

2. Explain the internal structure of leaf.

The internal structure of a leaf consists of: the upper epidermis, a protective layer covered with cuticle on the upper side of the leaf, mainly involved in water conservation; the palisade mesophyll just beneath the upper epidermis (regularly arranged elongated cells rich in chloroplasts, the main site of photosynthesis); the spongy mesophyll near the lower epidermis (irregularly shaped cells with air spaces among them to facilitate exchange of gases); the vascular bundles (xylem supplying water and minerals for

photosynthesis, phloem transporting food to all parts of the plant); and the lower epidermis, a protective layer on the lower side of the leaf having many pores called stomata for exchange of gases and removal of extra water, each stoma consisting of two guard cells that control its opening and closing.

3. Describe the adaptations in the structure of leaf that facilitate photosynthesis.

The structure of the leaf is well adapted to facilitate photosynthesis: (1) Large surface area — leaves are usually broad and flat, providing a large surface area exposed for absorption of sufficient sunlight intensity needed for photosynthesis. (2) Thin and transparent epidermis — the outer single layer of the leaf is thin and transparent, allowing sunlight to penetrate deeper into the leaf, reaching the cells where photosynthesis takes place. (3) Chloroplasts in mesophyll cells — mesophyll cells are rich in chloroplasts containing chlorophyll, which captures sunlight energy; the palisade mesophyll just below the upper epidermis and the spongy mesophyll next to the lower epidermis hold the chlorophyll. (4) Stomata for gaseous exchange — the maximum number of stomata in the lower epidermis facilitates efficient gaseous exchange and removal of extra water; stomata allow carbon dioxide to enter the leaf for photosynthesis and oxygen to exit as a by-product. (5) Network of veins spread throughout the leaf — the network of veins containing vascular bundles (xylem and phloem) is spread throughout the leaf; xylem supplies water and nutrients while phloem transports food prepared in the leaf to other parts of the plant body.

4. Give a comparative account between photosynthesis and respiration in plants.

Photosynthesis occurs in green plants, algae and some bacteria, while respiration takes place in all the living organisms. Photosynthesis takes place in chloroplasts in plant cells, while respiration takes place in mitochondria of all the cells in animals and plants. Photosynthesis uses sunlight energy to prepare food, while respiration releases energy from food. The products of photosynthesis are the reactants in respiration, and the products of respiration are the reactants in photosynthesis. Photosynthesis: Carbon dioxide + Water → (Sunlight, Chlorophyll) Food (Glucose) + Oxygen. Respiration: Food (Glucose) + Oxygen → Carbon dioxide + Water + Energy (ATP).

5. Describe the factors that affect transpiration.

Temperature: higher temperature increases the rate of transpiration because warmer air can hold more moisture. Humidity: higher humidity reduces the rate of transpiration because the air is already moist, so there is less difference in water potential between the leaf and the air. Wind: wind removes the moist air that accumulates around the leaf, creating a gradient for more water to evaporate, so wind increases the rate. Light intensity: higher light intensity increases the rate of transpiration because it stimulates the opening of stomata for photosynthesis, leading to more water loss.

6. Describe the applications of natural raise of water.

Natural raise of water is the phenomenon where water moves upward in plants from absorption through the roots to the leaves, and even beyond, using the process of transpiration — it is like a plant's own water pump. Its applications are: (1) Water rise helps transport nutrients dissolved in water from the soil up to the cells, keeping them firm and upright. (2) Transpiration and water rise help to cool the plant by releasing water vapours into the air — just like sweating cools our bodies, transpiration cools down the plant. (3) Water rise ensures that the leaves have a constant supply of water for photosynthesis. (4) Transpiration helps in the uptake of minerals and nutrients from the soil into the plant through the roots, and helps maintain turgor pressure, which keeps plant cells firm and upright.

Exercise (p.22): 1.5 Constructed Response Questions.

1. Describe briefly: (a) Principle of transpiration, (b) Plant's own water pump.

(a) Principle of transpiration: as water evaporates from the leaves it creates a 'pull' or suction force, similar to sucking liquid up a straw; this pull helps draw more water from the roots up through the plant. Water molecules attract together due to the intermolecular force called cohesive force (cohesion), and also stick to the walls of the xylem tubes due to adhesive force (adhesion). Liquids in narrow tubes (for example water in xylem) begin to move upwards due to adhesive force; the cohesive force helps water molecules move upwards in a continuous column, from the roots all the way to the leaves. (b) Plant's own water pump: natural water raise is a phenomenon where water moves upward by plants from absorption

through the roots to the leaves, and even beyond, using the process of transpiration. Transpiration creates a pull on water in the xylem, helping to transport water and nutrients from the roots to the leaves and the rest of the plant — so transpiration itself works as the plant's own water pump, with no mechanical pump involved.

2. Identify the differences between the following structures (a) and (b).

Structure (a) is the cross-section of a stem (the unit's Figure 1.5, internal structure of stem): it is a circular section whose labelled parts are the epidermis with root hairs on the outside, the cortex, the endodermis, the pericycle, the vascular cambium, and the primary xylem and primary phloem in the centre. Structure (b) is the cross-section of a leaf (the unit's Figure 1.8, internal structure of leaf): it is a flat section whose labelled parts are the cuticle and upper epidermis on top, the spongy mesophyll in the middle, the xylem and phloem of a vein, the stoma, and the lower epidermis at the bottom (the palisade mesophyll layer is shown between the upper epidermis and the spongy mesophyll). Differences: (a) is circular and built for conduction and growth in diameter — it has a cortex, an endodermis, a pericycle and a vascular cambium that produces new vascular tissue, and its outer layer bears root hairs; (b) is flat and built for photosynthesis and gaseous exchange — it has two epidermal layers with a cuticle on top, mesophyll tissue (palisade and spongy) between them, and stomata in the lower epidermis for exchange of gases and removal of extra water, which (a) does not have. In short, (a) shows how a stem conducts water and food and grows thicker, while (b) shows how a leaf captures

sunlight, makes food and exchanges gases.

Note: 1.6 Investigation and 1.7 Project (p.23) are marked 'Teacher Guide' in the book — the teacher facilitates students to investigate (what enables xylem, phloem, stomata, root hair and palisade mesophyll to perform their functions; why carbon dioxide and oxygen percentages in air remain constant) and to work out a project (demonstrate the rise of water based on the principle of transpiration using low-cost or locally available material). These are classroom investigations with no single written answer in the book, so no model answer is given for them here.

Unit 2: Human Respiratory and Circulatory Systems

Exercise (p.39): Multiple Choice Questions — 2.1 Encircle the correct option.

(i) The organ which controls the entrance of food and air in the relevant tubes is called:

- Throat
- Larynx
- ✓ Epiglottis
- Trachea

(ii) Aerobic respiration takes place in the cells in:

- ✓ Mitochondria
- Nucleus
- Cell membrane
- Vacuoles

(iii) Anaerobic respiration takes place in the cells in:

- Nucleus
- Cell membrane
- ✓ Cytoplasm
- Vacuoles

(iv) Byproducts of anaerobic respiration in yeast are:

- Carbon dioxide and water

✓ Carbon dioxide and alcohol

Oxygen and water

Alcohol and water

(v) The largest blood vessel that takes blood away from the heart to body parts is:

Pulmonary artery

Blood capillary

✓ Aorta

Pulmonary vein

(vi) Deoxygenated blood means:

Oxygen-rich blood

✓ Oxygen-poor blood

Blood present in left atrium

Blood present in left ventricle

(vii) Blood is carried from lungs to heart by:

Renal arteries

Pulmonary arteries

✓ Pulmonary veins

Renal veins

(viii) The narrowest blood vessels in human body are called:

Arteries

Veins

✓ Capillaries

Pulmonary arteries

(ix) Exchange of gases takes place in:

- Nose
- Nostrils
- Trachea
- ✓ Alveoli

(x) Exchange of material in the body tissue and the blood takes place in:

- Veins
- ✓ Capillaries
- Arteries
- Aorta

Exercise (p.40): 2.2 Write short answers.

1. What is meant by circulatory system?

The circulatory system is the system that carries blood through the body — it consists of the heart, the blood vessels, arteries and veins. The heart works like a pump that pushes blood in the blood vessels, and the vessels carry blood, with its oxygen, food, hormones and wastes, to and from every part of the body.

2. Define respiration.

Respiration is a biochemical process taking place in every cell of our body and all other organisms, in which oxygen reacts with food (glucose) to release energy in the form of ATP (Adenosine triphosphate). It happens inside the cells, specifically in tiny structures called mitochondria, and each of its reactions is

facilitated by a specific enzyme.

3. Where does blood oxygenate in the body?

Blood oxygenates in the lungs — oxygen of the air is absorbed in the blood capillaries after passing through the alveoli. In each lung the bronchus divides into small tubes called bronchioles, and at the end of each bronchiole there are extremely small air sacs called alveoli; a large number of blood capillaries are present in the alveoli, and oxygen from the air diffuses across the thin walls of the alveoli into the bloodstream.

4. State the function of red blood cells (RBCs).

Red blood cells (RBCs) are the most abundant cells in the blood. They contain a protein called haemoglobin, which binds to oxygen in the lungs and carries it to tissues throughout the body. They give blood its red colour and carry oxygen to the tissues while removing carbon dioxide from the tissues to the lungs for exhalation.

5. Write the names of different blood cells.

The different blood cells (components of blood) are: red blood cells (RBCs), white blood cells (WBCs), platelets, and plasma — the liquid component of blood, making up about 55% of its volume.

Exercise (p.40): 2.3 Differentiate between the following.

1. Oxygenated and deoxygenated blood

Oxygenated blood is oxygen-rich blood — it comes from the lungs and the heart pushes it to all body parts (the left ventricle pushes oxygenated blood into the aorta). Deoxygenated blood is

oxygen-poor blood — it is pushed by the heart towards the lungs (the pulmonary arteries carry deoxygenated blood from the right ventricle to the lungs, and all veins carry deoxygenated blood except the pulmonary veins, which bring oxygenated blood from the lungs to the left atrium).

2. Atria and ventricles

Atria are the upper two chambers of the heart; their walls are thin as compared to the ventricles, and each atrium opens into its ventricle (the right atrium opens in the right ventricle, the left atrium in the left ventricle). Ventricles are the lower two chambers; they are larger in size than the atria, their walls are thicker, and blood is pushed towards all body parts during pumping of the ventricles, while blood is pushed towards the ventricles during pumping of the atria. The heart works like a double pump — pumping of atria and pumping of ventricles.

3. Arteries and veins

Arteries are the blood vessels that carry blood from the heart to body parts; their walls are thick and flexible (thick, muscular walls with elastic fibres to withstand the high blood pressure pumped by the heart), and the largest artery is the aorta. Veins are the blood vessels that bring blood from various body parts back to the heart; their walls are thinner compared to arteries, and they contain valves that prevent blood from flowing backward and help maintain blood flow toward the heart. All veins carry deoxygenated blood except the pulmonary veins, which bring oxygenated blood from the lungs to the left atrium.

4. Capillaries and alveoli

Capillaries are the extremely narrow vessels into which arteries divide; they are the narrowest blood vessels in the body, with thin walls of only one layer of endothelial cells, and they form an extensive network throughout the body where oxygen and food in the blood diffuse into the body cells and carbon dioxide, excess water and waste materials from the cells diffuse into the blood. Alveoli are tiny, balloon-like air sacs clustered at the end of the bronchioles within the lungs; their main function is to facilitate the exchange of oxygen and carbon dioxide between the air and the bloodstream, and they are surrounded by a network of capillaries — oxygen diffuses across the thin walls of the alveoli into the bloodstream, while carbon dioxide diffuses from the bloodstream into the alveoli to be exhaled.

5. Aerobic and anaerobic respiration

Aerobic respiration is the process during which food is broken down to release energy by reaction in the presence of oxygen; it occurs inside the mitochondria, produces more energy, and its byproducts are carbon dioxide and water. Anaerobic respiration is the breakdown (decomposition) of food molecules to release energy in the absence of oxygen; it occurs mostly in the cytoplasm, releases less energy as compared to aerobic respiration, and its byproducts vary depending on the organism — in yeast they are alcohol and carbon dioxide (alcoholic fermentation), and in our muscles the byproduct is lactic acid, which does not produce carbon dioxide.

Exercise (p.40): 2.4 Constructed Response Questions.

1. When does anaerobic respiration happen in our body?

Anaerobic respiration happens in our body when enough oxygen is not available — it is like a backup plan of our bodies. It happens when we are doing intense exercise and our muscles cannot get enough oxygen; the breakdown of food molecules then releases energy in the absence of oxygen, producing less energy than aerobic respiration, and the byproduct in our muscles is lactic acid, which is transported to the liver, which converts it back into glucose.

2. When we do exercise, our heartbeat increases. Why?

When we do exercise, our heartbeat increases because the body needs more oxygen and energy during intense exercise, so the heart has to beat faster to deliver oxygen-rich blood to the muscles. The unit's hypothesis states it exactly: 'If the intensity of exercise increases, then the pulse rate will also increase. This is because the body needs more oxygen and energy during intense exercise, so the heart has to beat faster to deliver oxygen-rich blood to the muscles.' A healthy person's heartbeat is 72–80 times per minute at rest, and the pulse rate rises with low, moderate and high-intensity exercise (walking, jogging or cycling, sprinting or jumping jacks).

3. Explain the mechanism of breathing.

Breathing is the process which takes place in our lungs; it involves inhaling of oxygen and exhaling of carbon dioxide, and it does not produce energy — its main purpose is to bring oxygen into our

bodies and remove carbon dioxide, and it is controlled by our brain. Mechanism: the lungs are protected inside the chest; ribs provide support to the chest walls, there are two sets of muscles between the ribs, and the dome-like sheet of muscles separating the inner chest cavity from the abdomen is called the diaphragm. Inhaling: when the middle muscles of the ribs contract, the ribs are pushed outward, the chest expands outward and the diaphragm turns into a flat shape; the volume of the inner cavity increases and air enters into the lungs. Exhaling: in the reverse process the ribs and diaphragm expand back, the volume of the inner cavity decreases, the chest contracts inward, and air is expelled out. Path of air: nose and nostrils (hair and mucus keep the air clean, moist and warm) → throat and larynx (which has two vocal cords that produce sound; the epiglottis covers the breathing tube when food enters the oesophagus so food does not enter the breathing tube) → trachea (rings of cartilage keep the tube open) → two bronchi, one entering each lung → bronchioles → alveoli, where gas exchange takes place.

4. Describe the structure and function of human heart.

Structure: the human heart is a muscular organ, approximately equal in size to one's fist, located on the left side of the chest and enclosed in a membrane called the pericardium. It has four chambers: the upper two chambers are called atria (singular atrium) and the lower two chambers are called ventricles; the ventricles are larger in size than the atria, and the walls of the atria are thin as compared to the ventricles. Each atrium opens into its ventricle, and when the atria contract the blood enters the ventricles but cannot return to the atria on contraction of the

ventricles. Function: the heart works like a pump that pushes blood in the blood vessels — it works like a double pump, i.e. pumping of atria and pumping of ventricles; blood is pushed towards the ventricles during pumping of the atria and towards all body parts during pumping of the ventricles. It pushes the oxygenated blood (oxygen-rich blood) coming from the lungs to all body parts and pushes the deoxygenated blood (oxygen-poor blood) towards the lungs. The heartbeat is the result of contraction and relaxation of heart muscles; the heartbeat of a healthy person is 72–80 times per minute, and the heartbeat is also called the pulse, which can be determined by placing a finger below the base of the thumb on the underside of the wrist.

5. Describe the composition of blood.

Blood is a vital fluid (connective tissue) that circulates throughout the body, delivering essential substances and removing waste products. It is composed of: (a) Red blood cells (RBCs) — the most abundant cells in the blood; they contain haemoglobin, which binds to oxygen in the lungs and carries it to tissues throughout the body, give blood its red colour, and remove carbon dioxide from the tissues to the lungs for exhalation. (b) White blood cells (WBCs) — the cells of the immune system; they help protect the body against infections and disease-causing organisms (such as bacteria, viruses and fungi), and also play a role in inflammation, allergic reactions and immune responses to foreign invaders. (c) Platelets — small cell fragments in the blood; they play a crucial role in blood clotting, which helps stop bleeding when a blood vessel is injured, and they release growth factors that promote tissue repair and

healing. (d) Plasma — the liquid component of blood, making up about 55% of its volume; it is mostly water but also contains proteins, electrolytes, hormones, nutrients, waste products and gases, serves as a medium for immune system components and antibodies, and plays a role in maintaining blood pressure and body temperature.

6. Explain the functions of the followings: a. Arteries, b. Veins, c. capillaries.

(a) Arteries: the blood vessels that carry blood from the heart to body parts; their walls are thick, muscular and elastic to withstand the high blood pressure pumped by the heart; the largest artery is the aorta, which the left ventricle pushes oxygenated blood into, and it divides into small arteries carrying oxygenated blood to various parts of the body (the pulmonary arteries are the exception — they carry deoxygenated blood from the right ventricle to the lungs). (b) Veins: the blood vessels that bring blood from various body parts back to the heart; their walls are thinner compared to arteries and they contain valves that prevent blood from flowing backward and help maintain blood flow toward the heart; the largest veins in the body are the superior and inferior vena cavae, which carry blood from the body back to the heart (all veins carry deoxygenated blood except the pulmonary veins, which bring oxygenated blood from the lungs to the left atrium). (c) Capillaries: the extremely narrow vessels into which arteries divide — the narrowest blood vessels in the body, with thin walls of only one layer of endothelial cells; they form an extensive network throughout the body, including within organs, muscles and tissues,

and they connect arteries to veins; they provide oxygenated blood to all cells of various body parts, and they facilitate the exchange of gases, nutrients and waste products between blood and tissues — oxygen and food in the blood of capillaries diffuse into the body cells, while carbon dioxide, excess water and waste materials from the cells diffuse into the blood of capillaries, which then combine to form the veins.

Note: 2.5 Project (p.40) is marked 'Teacher Guide' in the book — the teacher facilitates students to make the model of the human heart using low-cost materials. It is a classroom project with no single written answer in the book, so no model answer is given for it here.

Unit 3: Immunity and Diseases

Exercise (p.57): Multiple Choice Questions — 3.1 Encircle the correct option.

(i) The ability of our body to fight against diseases is called:

- Enzyme specificity
- Chemical reactivity
- ✓ Immunity
- Scarcity

(ii) Salivary glands are present in the walls of:

- Intestine
- Stomach
- ✓ Oral cavity
- Nasal cavity

(iii) White blood cells that develop defense system are called:

- ✓ Lymphocytes
- Antibodies
- Antigens
- Pathogens

(iv) Body tissues on getting damaged by some invader, releases chemicals:

- Insulin

✓ Cytokines

Pepsin

Lysozyme

(v) Malaria is caused by:

Ascaris

Giardia

✓ Plasmodium

Entamoeba

(vi) Tuberculosis is caused by:

Staphylococcus

✓ Mycobacterium

Schistosoma

E. coli

(vii) Old and damaged red blood cells and pathogens are filtered out from the body by:

Appendix

Tonsils

✓ Spleen

Alveoli

(viii) Typhoid fever is caused by:

Taenia solium

Streptococcus

Candida

✓ Salmonella typhi

Exercise (p.58): 3.2 Define the following.

1. infection

Infection is the entry and growth of microorganisms (pathogens) into the body of another organism. Humans and all other living organisms are always exposed to the environment, and the air we breathe in, the water we drink and even the food we eat are not free of microorganisms and pathogens.

2. antibodies

Antibodies are proteins that protect the body against disease-causing organisms: they identify pathogens for their destruction by other immune cells and prevent them from infecting body cells.

3. antigens

Antigens are the foreign substances that enter our body — pathogens like viruses, bacteria and fungi, as well as allergens and toxins. The immune system responds to antigens by producing antibodies.

4. immunity

Immunity is the natural ability of the body that fights against diseases and keeps us safe. Whenever our body immunity against a certain disease becomes weak, the pathogens of that disease may attack us.

Exercise (p.58): 3.3 Differentiate between the following.

1. Bacterium and virus

Bacteria are single-celled microorganisms that are found almost everywhere, including soil, water and the human body; the majority of bacteria are harmless or even beneficial, but some bacteria cause diseases when they invade the body and multiply (for example Streptococcus causes strep throat, E. coli causes food poisoning, Staphylococcus aureus causes skin infections, and Mycobacterium tuberculosis causes tuberculosis). Viruses are tiny infectious agents that can only replicate inside the living cells of other organisms; they are much smaller than bacteria and cannot survive or reproduce outside a host cell (for example influenza virus causes flu, HIV causes AIDS, hepatitis viruses cause liver inflammation and disease, and rhinovirus causes the common cold).

2. Fungus and worm

Fungi are a group of organisms that include yeasts, molds and mushrooms; many fungi are harmless, but some can cause infections in humans, particularly in individuals with a weakened immune system (for example Candida albicans causes yeast infections, Trichophyton causes athlete's foot and ringworm, and Epidermophyton floccosum causes skin and nail infections). Worms (helminths) are multicellular parasitic organisms that infect humans and animals and include roundworms, tapeworms and flukes (for example Ascaris lumbricoides affects the intestine and causes ascariasis, Taenia solium causes intestinal tapeworm infection, and Schistosoma may affect the liver, bladder, brain and lungs, causing

schistosomiasis).

3. Vaccine and cytokines

A vaccine is a preparation of dead or weak pathogens of a certain disease injected into the body in order to produce active immunity — the body then produces its own antibodies in response to the pathogens, which takes time. Cytokines are chemicals released by body tissues when they get damaged by some invader; they cause inflammation — burning effect, redness, swelling and pain are the signs of inflammation — and the inflammatory response is a part of the body's innate (second-line) defence in which special inflammatory cells are created and sent to the affected area through blood flow to fight pathogens and promote tissue repair.

4. T-lymphocytes and B-lymphocytes

T-lymphocytes and B-lymphocytes are both lymphocytes — the white blood cells produced by the lymphoid organs — and both belong to the adaptive (third-line) immune response, which is specific and targeted, develops after exposure to specific pathogens or antigens, and provides long-term protection because memory cells formed during the first encounter mount a rapid and effective response upon re-exposure. Within that response, B-lymphocytes are the cells associated with the production of antibodies (the proteins that identify pathogens for destruction), while T-lymphocytes are the cells that directly recognise and destroy infected cells and coordinate the adaptive response.

5. Infection and immunity

Infection is the entry and growth of disease-causing microorganisms (pathogens) into the body. Immunity is the body's natural ability to fight against those diseases and keep us safe — its first line of defence is physical and chemical barriers like the skin and mucous membranes, its second line is the innate immune response (inflammation, phagocytes and natural killer cells), and its third line is the adaptive immune response (T cells, B cells and memory cells providing long-term immunity against specific pathogens).

Exercise (p.58): Constructed Response Questions.

1. Propose two strategies for strengthening our immune system.

Two strategies for strengthening our immune system (from the unit's common strategies) are: (1) Healthy diet — consume a balanced diet that includes a variety of fruits, vegetables, whole grains, lean proteins and healthy fats; each food group provides essential nutrients, vitamins and minerals that support immune function. (2) Regular exercise — participate in regular physical activities such as walking, jogging, cycling or playing sports; exercise boosts circulation of blood, promotes the production of immune cells, and reduces stress, all of which can enhance immune function. The unit also lists staying hydrated (water helps flush out toxins and maintains optimal bodily functions) and sufficient sleep (a consistent sleep schedule for optimal immune function and overall health).

2. Suggest two ways for safeguard against infectious diseases?

Two ways to safeguard against infectious diseases (from the unit's

additional layers of defence and safeguards) are: (1) Wearing masks — the mask acts as a barrier that helps prevent respiratory droplets, which may contain viruses or bacteria, from entering your nose and mouth, and it helps protect both you and others around you from spreading respiratory illnesses such as COVID-19 or the flu. (2) Practising good hygiene — regular hand washing for 20 seconds with soap and water is one of the most effective ways to prevent the spread of infectious diseases; it removes dirt, germs and viruses from your hands, and hand sanitizers help in killing germs on your hands and reduce the risk of transmitting pathogens to your mouth, nose or eyes. The unit also lists physical distancing (at least 1 metre / 3 feet), respiratory etiquette, vaccination and cleaning and disinfecting frequently touched surfaces.

3. Briefly describe lymphatic system.

The lymphatic immune system is one of the parts of the immune system (alongside the physical and chemical barriers, the innate immune system and the adaptive immune system). The organs involved in the immune system work to produce lymphocytes and are thus called lymphoid organs: the tonsils and adenoids, the thymus, the bone marrow, the axillary lymph nodes, the spleen, Peyer's patch, the appendix and the inguinal lymph nodes. Lymph nodes are small organs that filter lymph — a fluid containing white blood cells and waste products — to remove pathogens and foreign particles; they contain immune cells that help fight infections and activate the immune response. The spleen filters blood and removes old or damaged red blood cells and pathogens, and it contains immune cells that help produce antibodies and fight

infections. Bone marrow produces billions of new blood cells every day. Tonsils are a pair of soft tissues inside the throat that fight against infections and swell up in response to infections. Adenoids are a patch of soft tissue located behind the nasal cavity that traps harmful bacteria or viruses we breathe in or swallow, and they are important infection fighters in young children.

4. Briefly describe the importance of bone marrow for strengthening our defense system.

Bone marrow is vital for strengthening our defence system because it produces billions of new blood cells every day and releases them into the blood. The stem cells in the bone marrow differentiate into different cell types — red blood cells, white blood cells and platelets — and the white blood cells (including the lymphocytes produced by the lymphoid organs) are the cells of the immune system that fight infections, produce antibodies and activate the immune response. Without this continuous supply of new blood cells from the bone marrow, the body could not maintain its defence system.

5. Discuss preventive measures for Hepatitis and Covid-19.

Hepatitis: it is caused by viruses (hepatitis A, B, C, D and E) that infect the liver, often spread through contaminated food or water or contact with infected blood or bodily fluids. It is diagnosed through blood tests that detect specific antibodies or antigens associated with the virus, and liver function tests may also be performed to assess liver damage and function. It can be prevented through vaccination (especially for hepatitis A and B), practising good hygiene and avoiding contaminated food and water. COVID-19: it is

caused by the novel coronavirus SARS-CoV-2, spreading primarily through respiratory droplets when an infected person coughs, sneezes or talks, and also by touching contaminated surfaces. It is diagnosed by the polymerase chain reaction (PCR) test, which detects the virus's genetic material in respiratory samples such as nasal swabs, and rapid antigen tests may also be used for screening, providing results within minutes. It can be prevented by wearing masks, maintaining social distancing, washing hands frequently, avoiding large gatherings and getting vaccinated.

Note: MCQ (ii) as printed labels its first two options both as 'b.' (Intestine, Stomach) followed by 'c.' Oral cavity and 'd.' Nasal cavity; the options are rendered here in printed order with the intended a/b/c/d labels, and Oral cavity (c) is ticked, since saliva is secreted by salivary glands in the oral cavity and is antiseptic. 3.5 Project (p.58) is marked 'Teacher Guide' — the teacher facilitates students to manage and conduct a seminar in school on how people can safeguard against the spread of infectious diseases. It is a classroom project with no single written answer in the book, so no model answer is given for it here.

Unit 4: Structure of an Atom

Exercise (p.69): Multiple Choice Questions — 4.1 Encircle the correct option.

(i) The lightest particle of atom is:

Nucleus

Proton

✓ Electron

Neutron

(ii) The maximum number of electrons in third shell (M) around the nucleus is?

8

10

✓ 18

32

(iii) The number of neutrons present in Helium atom are:

1

✓ 2

3

4

(iv) An element with ten protons in its atomic nucleus has atomic number:

- ✓ 10
- 11
- 12
- 20

(v) An element with atomic number 4 has number of electrons around the nucleus of its one atom:

- 2
- ✓ 4
- 6
- 8

(vi) The shell represented by letter 'L' is:

- 1st
- ✓ 2nd
- 3rd
- 4th

(vii) The central part of an atom is:

- Electrons
- Protons
- Neutrons
- ✓ Nucleus

(viii) A neutral particle in the atom is:

- Electron
- Proton

✓ Neutron

Nucleus

(ix) Atomic number of fluorine is 9. The number of protons in it is:

3

6

✓ 9

13

(x) The horizontal rows in the Periodic Table are called:

Groups

✓ Periods

Classes

Families

Exercise (p.70): 4.2 Differentiate between the following.

1. Proton and electron

A proton is a positively charged particle of an atom found in its nucleus; it is 1836 times heavier than an electron. An electron is a negatively charged particle of an atom found around its nucleus; it is the lightest particle of an atom and revolves around the nucleus in shells (orbits or energy levels).

2. Atomic number and mass number

Atomic number (Z) is the number of protons present in the nucleus of an atom of an element; under normal conditions the number of protons is always equal to the number of electrons in any atom. Mass number (A) is the sum of the number of protons and neutrons

present in the nucleus of any atom, given by the formula A = Z + n, where n denotes the number of neutrons (for example, hydrogen has one proton and no neutron, so its mass number is 1; helium has two protons and two neutrons, so its mass number is 4).

3. Group and period

Groups are the vertical columns in the Periodic Table; there are eighteen groups, and elements kept in the same column (group) have the same number of electrons in their outermost (valence) shells and similar properties — groups are numbered on the basis of the number of electrons in the outermost shells (for example, each element of group 1 contains one electron in its outermost shell, and each element of group 2 contains two). Periods are the horizontal rows in the Periodic Table; there are seven periods, and elements in the same period have the same number of atomic orbitals — in the rows (periods) from left to right, elements are arranged in order of increasing atomic numbers (for example, the 1st period is the shortest period with atomic numbers 1 to 2, the 2nd and 3rd are short periods of 8 elements each, and the 4th and 5th are long periods of 18 elements each).

4. K shell and M shell

The K shell is the first shell around the nucleus (position n = 1); by the formula for the maximum number of electrons in a shell, $2n^2$, it holds at most $2 \times 1^2 = 2$ electrons. The M shell is the third shell around the nucleus (position n = 3); by the same formula it holds at most $2 \times 3^2 = 18$ electrons. The second shell, L (n = 2), holds at most $2 \times 2^2 = 8$ electrons.

Exercise (p.70): 4.3 Constructed Response Questions.

1. Describe the structure of an atom.

An atom consists of three fundamental particles: electron, proton and neutron. Protons and neutrons are found in the central part of an atom that is called the nucleus, and almost the whole mass of the atom is due to protons and neutrons — as these two particles are found in the nucleus, the whole mass of the atom is the mass of the nucleus. A proton has a positive charge and is 1836 times heavier than an electron. A neutron is a neutral particle, i.e. it has no charge; it is 1842 times heavier than an electron and has almost the same mass as that of a proton. An electron is the lightest particle of an atom; it has a negative charge and revolves around the nucleus in shells, also called orbits or energy levels. The number of electrons in an atom is always equal to the number of protons in it, so an atom is electrically neutral: equal quantities of positive and negative charges cancel the effect of each other. The smallest atom is the hydrogen atom: it has one proton and one electron, and the electron revolves around the nucleus in the first orbit or shell, and the hydrogen atom has no neutron. A helium atom contains two protons, two neutrons and two electrons. The first, second and third shells are represented by K, L and M respectively, with maximum electron capacities of 2, 8 and 18 by the formula $2n^2$.

2. Draw atomic structure diagrams for the following: (a) Helium (b) Boron (c) Oxygen (d) Lithium (e) Carbon (f) Chlorine

Atomic structure diagrams (nucleus = protons + neutrons;

electrons arranged in shells K, L, M from the unit's Tables 4.1 and 4.3): (a) Helium (He) — atomic number Z = 2, mass number A = 4: nucleus with 2 protons and 2 neutrons; 2 electrons in the K shell. (b) Boron (B) — Z = 5, A = 11: nucleus with 5 protons and 6 neutrons; 2 electrons in the K shell and 3 electrons in the L shell. (c) Oxygen (O) — Z = 8, A = 16: nucleus with 8 protons and 8 neutrons; 2 electrons in the K shell and 6 electrons in the L shell. (d) Lithium (Li) — Z = 3, A = 7: nucleus with 3 protons and 4 neutrons; 2 electrons in the K shell and 1 electron in the L shell. (e) Carbon (C) — Z = 6, A = 12: nucleus with 6 protons and 6 neutrons; 2 electrons in the K shell and 4 electrons in the L shell. (f) Chlorine (Cl) — Z = 17, A = 35.5 as printed in the unit (17 protons and 18 neutrons in Table 4.3): nucleus with 17 protons and 18 neutrons; 2 electrons in the K shell, 8 electrons in the L shell and 7 electrons in the M shell. In each diagram the number of protons equals the number of electrons, and the number of neutrons is either equal to the number of protons (or electrons) or greater than that number.

Note: MCQ (vi) as printed labels its first two options both as 'b.' (1st, 2nd) followed by 'c.' 3rd and 'd.' 4th; the options are rendered here in printed order with the intended a/b/c/d labels, and 2nd (b) is ticked, since the first, second and third shells are represented by K, L and M respectively. In exercise 4.3 question 2 the book prints the items in the order (a) Helium, (b) Boron, (c) Oxygen, (d) Lithium, (f) Carbon, (e) Chlorine — Carbon labelled (f) and Chlorine labelled (e); they are answered here in the same item order with the print labels noted. 4.4 Project (p.70) is marked 'Teacher Guide' — the teacher facilitates students to conduct a discussion on the periodic table in

a group activity. It is a classroom activity with no single written answer in the book, so no model answer is given for it here.

Unit 5: Physical and Chemical Changes

Exercise (p.84): Multiple Choice Questions — 5.1 Encircle the correct option.

(i) An example of physical change is:

- Formation of curd from milk
- Decay of fruits
- ✓ Formation of clouds
- Production of energy from fuel

(ii) An example of chemical change is:

- Boiling of water
- ✓ Production of energy from food
- Dissolution of salt in water
- Freezing of water

(iii) The gas produced by reaction of zinc metal with hydrochloric acid is:

- ✓ Hydrogen
- Carbon dioxide
- Methane
- Oxygen

(iv) CH_4 + ... → CO_2 + Heat (the missing reactant is:)

- N_2
- H_2

- ✓ O2
- I2

(v) Fe + ... → Fe2O3 (the missing reactant is:)

- O2 (dry)
- ✓ O2 (moist)
- H2 (dry)
- N2 (moist)

(vi) Freezing of a liquid is a:

- Chemical change
- Chemical property
- ✓ Physical change
- None of above

(vii) A piece of iron is kept in open air for 5 days. A film of corrosion formed over it is:

- ✓ Iron oxide
- Iron sulphide
- Iron chloride
- Iron hydride

(viii) During combustion, a substance reacts with:

- Hydrogen
- Water
- Carbon dioxide
- ✓ Oxygen

Exercise (p.85): 5.2 Write short answers.

1. What type of change occurs when water is converted into steam?

A physical change — water is converted into steam by evaporation, and water maintains its chemical structure (H2O) and chemical properties in all three forms: ice (solid), water (liquid) and steam (gas).

2. What type of change occurs when ice melts into liquid water?

A physical change — melting of ice into liquid water changes only the state (form or appearance) of the substance without changing its chemical structure and properties, and it can be reversed easily by freezing.

3. What type of change occurs when water is converted into hydrogen gas and oxygen gas?

A chemical change — the chemical structure of the substance is changed and it is transformed into new substances: water is decomposed into hydrogen gas and oxygen gas, which are chemically quite different substances from water and cannot be converted back into water by physical methods.

4. Name the pollutants causing acid rain.

Sulfur dioxide (SO2), nitric oxide (NO) and volatile organic compounds (VOCs) — the unit's Table 5.1 lists these as the combustion emissions causing smog and acid rain.

5. What type of change occurs when chemical structure of a substance is changed?

A chemical change — chemical changes are those changes in which the chemical composition or chemical structure of substances are changed and they are transformed into new substances.

6. What are the reactants when we burn methane?

Methane (CH4) and oxygen (O2) — combustion is a chemical reaction between a fuel like natural gas (CH4) and oxygen, and oxygen is essential for combustion to occur because it acts as a reactant in the chemical reaction.

7. What are the products when we burn methane?

Carbon dioxide (CO2) and water (H2O), along with heat and light — during combustion, oxygen molecules (O2) combine with the fuel molecules to produce heat energy and new substances, typically carbon dioxide and water.

Exercise (p.85): 5.3 Differentiate between the following.

1. Ice and vapours

Ice is the solid form of water (H2O), formed when water freezes at 0°C; vapours (steam) are the gaseous form of water (H2O), formed when water boils at 100°C or evaporates. Both are the same substance chemically — water maintains its chemical structure (H2O) and chemical properties in all three forms — but they differ in state and physical properties: ice is a solid that melts on warming, while vapours are a gas that condenses on cooling.

2. Flammability and conductivity

Flammability is a chemical property: it is the ability of a substance

to catch fire (to burn) — for example, burning of acetylene with oxygen produces a high-temperature flame used for welding and cutting of metals, and materials are used on the basis of this property in preparing useful products. Conductivity is a physical property: thermal conductivity is the ability of a material to conduct heat (materials with high thermal conductivity, such as metals like copper and aluminum, are used in heat exchangers, electrical wiring and cooking utensils, while materials with low thermal conductivity, such as wood, polystyrene, wool and plastic, are used in insulation and packaging), and electrical conductivity is the physical property of metals to conduct electric current (for example, copper wire is used in electric wirings in our homes, schools and industries).

3. Rusting and tarnishing

Rusting is a chemical process in which iron reacts with oxygen and moisture in the air to form iron oxide (rust) on the surface of the metal; oxygen is a key reactant because it combines with iron atoms to form iron oxide, and the presence of moisture (water) is also necessary because it facilitates the chemical reaction between iron and oxygen. Tarnishing is a chemical process in which metals such as silver, copper or brass react with oxygen and sulfur compounds in the air to form a dark, dull layer on the surface — for example, silver reacts with hydrogen sulfide (H2S) and oxygen to form silver sulfide (Ag2S) and water, giving tarnished coins their dark, dull layer. The unit's vocabulary defines tarnish as a thin film of corrosion formed over the objects of copper, brass, magnesium, etc.

4. Chemical change and chemical equation

A chemical change is a change in which the chemical composition or chemical structure of substances are changed and they are transformed into new substances (for example, burning of objects, rusting of iron, formation of curd from milk, spoiling of milk, digestion of food, photosynthesis and respiration). A chemical equation is the representation of a chemical change or chemical reaction in terms of symbols, formulae and signs: the formulae or symbols of the reactants are written on the left side and the formulae or symbols of the products are written on the right side, separated by an arrow directed towards the products (for example, sulfur combines with oxygen to form sulfur dioxide: S (s) + O2 (g) → SO2 (g); and zinc with hydrochloric acid: Zn (s) + 2HCl (aq) → ZnCl2 (aq) + H2 (g), where (s) indicates the solid state, (aq) the aqueous state and (g) the gaseous state).

5. Physical properties and chemical properties

Physical properties are due to the physical behaviour of the substances and include hardness (ability to withstand scratches and wear — an iron nail scratches plastic, diamond is extremely hard and used to cut other materials, graphite is soft and leaves marks on paper), melting point (temperature at which a solid changes into its liquid state) and boiling point (temperature at which a liquid changes into its gaseous state), solubility (the maximum amount of a substance that can be dissolved in 100 gram of a solvent at a particular temperature and pressure — water is the most common solvent), and thermal conductivity (ability to

conduct heat). Physical changes only affect physical properties without changing chemical composition. Chemical properties are associated with the chemical structure of the substance and include reactivity (tendency of a substance to undergo chemical reactions — for example, decomposition of baking powder on heating produces carbon dioxide, which makes bakery products porous, soft and fluffy) and flammability (ability of substances to burn); a change in the chemical properties of a substance is called a chemical change or chemical reaction.

Exercise (p.85): 5.4 Constructed Response Questions.

1. Wet clothes are hanged in the sun and they become dry. (a) What happens to the water in wet clothes? (b) What makes its removal from the clothes? (c) Is this a chemical change? (d) Name the process involved in this change.

(a) The water in the wet clothes evaporates — it changes from liquid water into water vapours, which pass into the air, and the clothes become dry. (b) The heat of the sun (sunlight) makes its removal: on warming, liquid water evaporates into vapours, just as the unit's introduction describes — when you spread wet clothes in sunlight, the water present in wet clothes evaporates after some time and the clothes become dry. (c) No, this is not a chemical change — it is a physical change, because physical changes do not change the chemical composition or the structure of a substance: water maintains its chemical structure (H_2O) in all its forms, and physical changes can be reversed easily. (d) The process involved in this change is evaporation.

Note: In MCQ (iv) the printed equation shows only 'CH4 + ... → CO2 + Heat'; the missing reactant O2 is ticked because combustion is the reaction of the fuel methane with oxygen, and the unit's full combustion equation gives carbon dioxide and water as products with heat. In MCQ (v) the printed equation shows only 'Fe + ... → Fe2O3'; O2 (moist) is ticked because rusting requires both oxygen and moisture — dry oxygen alone does not rust iron. 5.5 Investigation and 5.6 Project (p.85) are both marked 'Teacher Guide' — the teacher facilitates students to conduct a discussion on the impact of combustion reactions on the environment, and on using physical and chemical properties of substances. They are classroom discussions with no single written answer in the book, so no model answer is given for them here.

Unit 6: Chemical Bonds

Exercise (p.97): Multiple Choice Questions — 6.1 Encircle the correct option.

(i) Two or more atoms combine to form:

- Anion
- Cation
- ✓ Molecule
- Radicals

(ii) One of the following is a cation:

- H_2
- H_2O
- CO_2
- ✓ NH_4^+

(iii) Which of the following is an anion?

- F
- Cl
- ✓ S^{2-}
- NH_3

(iv) Oxygen is a:

- Monovalent element
- ✓ Divalent element
- Trivalent element

Tetravalent element

(v) Chemical bond between N molecule is:

Ionic

Single covalent

Double covalent

✓ Triple covalent

(vi) How many atoms form one molecule of chlorine gas?

1

✓ 2

3

4

(vii) How many electrons are present in the outermost shell of sodium atom?

✓ 1

2

3

4

(viii) How many electrons are present in the outermost shell of chlorine atom?

1

2

6

✓ 7

(ix) Element with the stable electronic configuration is:

H

✓ He

C

Li

(x) Nitrogen atom has electrons in their valence shell:

One

Three

✓ Five

Two

Exercise (p.98): 6.2 Define the following and explain with examples.

1. Valency

Valency is the combining capacity of an atom of the element — the number of electrons that one atom of an element loses, gains or provides for sharing during chemical combination of elements. For example, during the formation of sodium chloride, the sodium atom acquires a noble-gas-like electronic configuration by losing one electron while the chlorine atom gains one electron, so the valency of both sodium and chlorine is one. For the formation of hydrogen chloride, the hydrogen atom shares one electron with the chlorine atom, so the valency of hydrogen is one. An oxygen atom shares two electrons with two atoms of hydrogen to form a water molecule, so the valency of oxygen is two. A nitrogen atom shares three electrons with three hydrogen atoms to form an ammonia molecule, so the valency of nitrogen is three. A carbon atom shares four electrons with four hydrogen atoms to form a methane

molecule, so the valency of carbon is four.

2. Ion

An ion is the charged particle resulting when an atom or group of atoms loses or gains electrons. With the loss of electron(s), a positive charge appears on that atom or group of atoms, and the particle with positive charge is called a cation (for example, a sodium atom loses one electron to form the sodium ion Na+, and a magnesium atom loses two electrons to form Mg2+). With the absorption of electron(s), there is an excess of negative charge on that atom or group of atoms, and the particle with negative charge is called an anion (for example, a chlorine atom gains one electron to form the chloride ion Cl-, and a sulfur atom gains two electrons to form the sulfide ion S2-). An anion composed of a single atom is called a simple ion, whereas an ion comprising more than one atom is called a molecular ion; ammonium ion (NH4+) and hydroxide ion (OH-) are examples of molecular ions.

3. Chemical formula

A chemical formula is the aggregate of symbols of atoms present in a molecule of any compound. It describes the types of elements and the number of atoms of each element present in a compound. For example, the formula for methane is CH4, which indicates that a methane molecule is composed of one carbon atom and four atoms of hydrogen, and the formula for water is H2O, which indicates that water is a compound consisting of two elements, hydrogen and oxygen, and one molecule of it has two atoms of hydrogen and one atom of oxygen. Other examples from the unit's

Table 6.2: ammonia NH_3 (N = 1, H = 3), glucose $C_6H_{12}O_6$ (C = 6, H = 12, O = 6), sulfuric acid H_2SO_4 (H = 2, S = 1, O = 4), sodium bicarbonate $NaHCO_3$ (Na = 1, H = 1, C = 1, O = 3), nitric acid HNO_3 (H = 1, N = 1, O = 3) and sucrose (sugar) $C_{12}H_{22}O_{11}$ (C = 12, H = 22, O = 11).

4. Bond

A chemical bond is the force that combines the atoms together — a force of attraction between atoms that holds them together in a substance. Atoms of the elements tend to get stability by gaining 2 or 8 electrons in their outermost shell like noble gases, and for this tendency atoms combine with other atoms, forming chemical bonds. Chemical bonds are formed in three ways: by donating valence shell electrons to other atoms, by gaining electrons from the valence shell of other atoms, and by mutual sharing of valence shell electrons between the atoms. Chemical bonds are of two types: ionic bond (formed by complete transfer of electron(s) — for example, sodium loses an electron and chlorine gains it to form sodium chloride, NaCl) and covalent bond (formed by mutual sharing of electrons between the atoms — single covalent bond by sharing one electron pair, e.g. H_2, Cl_2, HCl, H_2O; double covalent bond by sharing two electron pairs, e.g. O_2, CO_2; triple covalent bond by sharing three electron pairs, e.g. N_2).

5. Electronic configuration

Electronic configuration is the arrangement of electrons around the nucleus of an atom. The electronic configuration of noble gases is the stable electronic configuration: atoms of the elements tend to

get stability by gaining 2 or 8 electrons in their outermost shell like noble gases. For example, a sodium atom has 1 electron in its outermost shell (configuration 2, 8, 1) and loses it to acquire a noble-gas-like configuration, a chlorine atom has 7 electrons in its outermost shell (2, 8, 7) and gains one electron for stability, a nitrogen atom has 5 electrons in its outermost shell (2, 5), and helium (He), with 2 electrons in its K shell, has a stable electronic configuration.

Exercise (p.98): 6.3 Answer the following questions.

1. Write the symbols for the following elements. Copper, iron, chlorine, potassium, argon, phosphorus, beryllium, carbon, calcium, silicon.

Copper — Cu; iron — Fe; chlorine — Cl; potassium — K; argon — Ar; phosphorus — P; beryllium — Be; carbon — C; calcium — Ca; silicon — Si.

2. Write the chemical formulae of the following compounds. Sodium hydrogen carbonate, calcium hydroxide, ammonia, potassium chloride, acetic acid, ammonium hydroxide, sulfuric acid, nitric acid, benzene, methane.

Sodium hydrogen carbonate — NaHCO3; calcium hydroxide — Ca(OH)2; ammonia — NH3; potassium chloride — KCl; acetic acid — CH3COOH; ammonium hydroxide — NH4OH; sulfuric acid — H2SO4; nitric acid — HNO3; benzene — C6H6; methane — CH4. (Ionic formulae follow the unit's method: write the cation on the left and the anion on the right, then exchange the valency numbers — for example, Ca2+ and OH- give Ca(OH)2, and K+ and Cl- give KCl;

NaHCO3, NH3, H2SO4 and HNO3 are given in the unit's Table 6.2 and CH4 in section 6.1.3.)

3. In what way the molecule of an element is different from that of a compound?

A molecule of an element is formed when two or more atoms of the same element combine — for example, a hydrogen molecule (H2), a chlorine molecule (Cl2), an oxygen molecule (O2) and a nitrogen molecule (N2) are all molecules of elements, since each contains only atoms of one element. A molecule of a compound is formed when atoms of one element combine chemically with atoms of another element — for example, a water molecule (H2O) contains hydrogen and oxygen atoms, a carbon dioxide molecule (CO2) contains carbon and oxygen atoms, an ammonia molecule (NH3) contains nitrogen and hydrogen atoms, and a methane molecule (CH4) contains carbon and hydrogen atoms. In short, the molecule of an element contains atoms of one element only, while the molecule of a compound contains atoms of two or more different elements chemically combined.

4. Differentiate the following: (a) Symbol and chemical formula. (b) Positive ion and negative ion.

(a) Symbol and chemical formula: a symbol is the short representation of a single element (for example, H for hydrogen, Na for sodium, Cl for chlorine), while a chemical formula is the aggregate of symbols of atoms present in a molecule of a compound, showing the types of elements and the number of atoms of each element in it (for example, H2O for water and CH4

for methane). (b) Positive ion and negative ion: a positive ion (cation) is formed when an atom or group of atoms loses electron(s) and acquires a positive charge (for example, Na+, Mg2+, NH4+); a negative ion (anion) is formed when an atom or group of atoms gains electron(s) and acquires a negative charge (for example, Cl-, S2-, OH-).

Note: MCQ (iii) as printed labels its options 'a. F, b. Cl, b. S2-, d. NH3' — the third option is mislabelled 'b.'; the options are rendered here in printed order with the intended a/b/c/d labels, and S2- (c) is ticked, since it is the only option carrying a negative charge (the sulfide ion, Table 6.1), while F and Cl are shown as neutral element symbols and NH3 is a neutral molecule. 6.4 'Investigate the metallic bond' (p.98): the unit teaches only two types of chemical bond — ionic bond and covalent bond — and does not define the metallic bond anywhere in Unit 6, so no model answer is given for this investigation here; it is left as a classroom investigation. 6.5 Project (p.98): construct models to explain ionic and covalent bonds using low cost or locally available material — a classroom project with no single written answer in the book, so no model answer is given for it here.

Unit 7: Solutions

Exercise (pp.107–108): Multiple Choice Questions — 7.1 Encircle the correct option.

(i) A solution in which no more solute can dissolve at a fixed temperature is called solution.

- Supersaturated
- Unsaturated
- ✓ Saturated
- Homogenous

(ii) The solubility of which compound in water decreases by increasing the temperature?

- Sodium chloride
- ✓ Carbon dioxide
- Copper sulphate
- Potassium nitrate

(iii) Air is a solution. What is solvent in this solution?

- ✓ Nitrogen
- Oxygen
- Carbon dioxide
- Water

(iv) Solubility of sodium chloride in water at room temperature and pressure is:

18 g NaCl

✓ 36 g NaCl

54 g NaCl

100 g NaCl

(v) Solution consisting of two substances is called - solution.

✓ Binary

Mixed

Heterogeneous

Homogenous

Exercise (p.108): 7.2 Complete the sentences using proper words.

(i) The part of the solution which is present comparatively in more quantity is called _____.

The part of the solution which is present comparatively in more quantity is called solvent.

(ii) An unsaturated solution can dissolve more amount of the _____.

An unsaturated solution can dissolve more amount of the solute.

(iii) The solubility of lime (calcium oxide) in water decreases by increasing the _____.

The solubility of lime (calcium oxide) in water decreases by increasing the temperature.

(iv) Perfume solutions are prepared by dissolving sweet-smelling oil in _____.

Perfume solutions are prepared by dissolving sweet-smelling oil in alcohol.

(v) The solutions in which water is used as solvent are called _____.

The solutions in which water is used as solvent are called aqueous solutions.

Exercise (p.108): 7.3 Match the statements of column A with the relevant statements of column B.

Alloy →

A solution in both the solute and solvent are solids (for example, brass and bronze).

Dilute solution →

Little amount of salt dissolved in a glass full of water.

Solubility →

The amount of solute in grams required to make its saturated solution in 100 gram solvent at a fixed temperature (column B prints this as: 'Saturated solution of a solute in 100 gram solvent at a fixed temperature').

Binary solution →

A solution consisting of two substances.

Exercise (p.109): 7.4 Write short answers.

(i) What is aqueous solution?

An aqueous solution is a solution in which water is used as solvent. Water is the most important solvent found on the Earth, and a lot of substances can be dissolved in it — that is why it is known as the universal solvent (however, grease, paints, oil, fat, etc., do not dissolve in water).

(ii) What is meant by the term solubility?

Solubility is the amount of solute in grams which is required to make its saturated solution in 100 gram solvent at a fixed temperature. For example, in order to make a saturated solution of common salt (sodium chloride) in 100 gram water at normal temperature (25°C), 36 gram of common salt is required, and for a saturated solution of copper sulphate at normal temperature (25°C), 24 gram of copper sulphate is needed.

(iii) Give an example of solution in which both the solute and the solvent are solid.

An alloy — a solution in which both the solute and the solvent are solids, for example brass and bronze.

(iv) Give an example of solution in which both the solute and the solvent are liquid.

Ink in water (or milk in water) — a solution in which both the solute and the solvent are liquids.

(v) Name the factor that affect the solubility.

The factors that affect solubility (section 7.3.1) are: the nature of the solute and solvent, the temperature of the solution, and

pressure. The unit's Key Points single out temperature: temperature affects the solubility of substances.

Exercise (p.109): 7.5 Differentiate between the following.

(i) Solute and solvent.

The solute is the part of the solution which is present in less (smaller) quantity, whereas the solvent is the part of the solution which is present in more (major, large) quantity. In a solution of sugar and water, sugar is the solute and water is the solvent; the solute and solvent are known as the components of the solution.

(ii) Solution and solubility.

A solution is a homogeneous mixture of two or more substances — its colour, density, appearance and other physical and chemical properties are the same in every part of the solution. Solubility is the amount of solute in grams which is required to make its saturated solution in 100 gram solvent at a fixed temperature.

(iii) Dilute solution and concentrated solution.

A dilute solution is a solution which has less quantity of dissolved solute in it (for example, a little amount of salt dissolved in a glass full of water, or two drops of ink in a beaker of water giving a light colour). A concentrated solution is a solution which has comparatively more quantity of dissolved solute in it (for example, four drops of ink in a beaker of water giving a deep colour).

(iv) Saturated solution and unsaturated solution.

An unsaturated solution is a solution in which, at a fixed

temperature, a further quantity of solute can be dissolved. A saturated solution is a solution in which, at a fixed temperature, no more quantity of solute can be dissolved — if sugar is added and stirred until further dissolution stops and insoluble sugar crystals settle down at the bottom of the beaker, the filtrate is a saturated solution.

Exercise (p.109): 7.6 Sea water is called solution. Why?

7.6 Sea water is called solution. Why?

Sea water is called a solution because it is a homogeneous mixture of two or more substances: water is the solvent and the salts dissolved in it are the solutes, mixed uniformly so that its composition is the same in every part. The unit's 'Do you Know?' box adds that the sea is highly saturated with salts (a supersaturated solution) — with a slight decrease in temperature, the salt crystals start appearing in it.

Exercise (p.109): 7.7 White washing.

7.7 For white washing the house, you have to dissolve more and more lime in water. Which of the hot water or cold water will you use for this purpose? And why?

Cold water should be used. The solubility of lime (calcium oxide) in water decreases by increasing the temperature of the solution, so more lime dissolves in cold water than in hot water — dissolving more and more lime for white washing is therefore done in cold water.

Exercise (p.109): 7.8 Constructed Response Questions.

(i) Describe preparation of a solution.

To prepare a solution, take a measured amount of solvent (for example, water in a glass or beaker) and add the solute to it in small amounts, stirring thoroughly until it dissolves uniformly. For example, when a tea spoon of sugar is added to a glass of water and stirred, the sugar dissolves in the water in such a way that a homogeneous mixture is formed. By the particle model, when a solute is mixed in a solvent, due to constant motion the particles of the solute and solvent interact with each other: the attractive forces between the particles of the solute are weakened and attractive forces between the solute and solvent particles are produced, so the solute particles begin to enter the intermolecular spaces of the solvent, and finally the particles of the solute and solvent are mixed up homogeneously to form a solution.

(ii) Describe the factors affecting the solubility.

The factors affecting solubility (section 7.3.1) are: (1) Nature of the solute and solvent — the principle of dissolution is that like dissolves like: water has ionic character, so the ionic compounds like sodium chloride, silver nitrate, silver chloride, etc., are dissolved in water, while covalent compounds are dissolved in covalent solvents. (2) Temperature of the solution — temperature affects the solubility of substances: the solubility of some compounds, e.g. sodium chloride (common salt), copper sulphate and potassium nitrate, etc., in water increases with the increase of temperature of the solution, while there are certain compounds, e.g. carbon dioxide and calcium oxide (lime), etc., whose solubility decreases by increasing the temperature of the solution (for

example, at room temperature 25°C the solubility of copper sulphate in water is 24 gram, whereas at 60°C its solubility becomes 60 gram). (3) Pressure — pressure affects the solubility of gaseous solutes in liquid solvents: carbon dioxide is dissolved in soda water, and when the cap on the soda bottle is opened, the pressure on the solution is released and the carbon dioxide dissolved under pressure begins to bubble out.

(iii) Explain the effect of temperature on the solubility of table salt in water.

The solubility of table salt (sodium chloride) in water increases with the increase of temperature. At normal temperature (25°C), 36 gram of common salt is required to make a saturated solution in 100 gram water, and at that temperature no more salt will dissolve in its saturated solution. But if the solution is heated and some further salt is added, some further amount of the salt dissolves in its saturated solution on heating — so heating a saturated salt solution raises its solubility and more salt can dissolve in it.

(iv) Describe the importance of solutions in daily life.

Solutions are important in daily life in many ways (section 'How does solubility affect everyday life?'): (1) In carbonated drinks, carbon dioxide remains dissolved in water due to its solubility in water. (2) Paints are dissolved in solvents like alcohol and ether, etc., and these solvents help to dry the paint. (3) Water-soluble dyes are used in the textile industry at large scale, as dyes (coloured substances) are used to create colourful patterns on clothes. (4) Solubility is very important in the pharmaceutical

industry: it is helpful to achieve the desired concentration of the drugs in solutions. (5) The use of solubility parameters is very important in the petroleum industry. Beyond this, water as the universal solvent dissolves the food we eat after digestion to form a solution which is absorbed in blood for reaching every part of our body, and nutrients in soil are dissolved in water for their diffusion into roots.

(v) How can we increase the process of dissolution?

The process of dissolving can be increased (unit's learning objectives) by: (1) Increasing the temperature — heating the solvent increases the solubility of most solids and the motion of the particles, so the solute dissolves faster and in greater amount. (2) Stirring — stirring brings fresh solvent into contact with the solute particles and spreads the dissolved particles through the solvent. (3) Breaking the solid into smaller pieces — crushing or powdering the solute increases its surface area exposed to the solvent, so it dissolves faster.

7.9 Investigate the importance of solubility in textile and pharmaceutical industries.

In the textile industry, water-soluble dyes are used at large scale, since dyes (coloured substances) are used to create colourful patterns on clothes — the solubility of the dye in water is what makes dyeing possible. In the pharmaceutical industry, solubility is very important because it is helpful to achieve the desired concentration of the drugs in solutions — a drug must dissolve to the required concentration in its solution form to be effective.

Note: MCQ (i) as printed labels its options 'a. Supersaturated, b. Unsaturated, b. Saturated, d. Homogenous' — the third option is mislabelled 'b.'; the options are rendered here in printed order with the intended a/b/c/d labels, and Saturated (c) is ticked. In MCQ (i) and (v) the book prints 'Homogenous' for 'Homogeneous'; the standard spelling is used in the answers. The Unit 7 opener (p.99) prints the unit number as '200' in the scanned layout (a design artefact); the unit is Unit 7, Solutions, as confirmed by the contents page and the exercise numbering 7.1–7.9.

Unit 8: Force and Motion

Exercise (pp.124–125): Multiple Choice Questions — 8.1 Encircle the correct option.

(i) The example of non-contact force is:

- Push
- Pull
- Friction
- ✓ Gravity

(ii) The SI unit of force is:

- ✓ Newton
- Metre per second
- Metre
- Kilometre

(iii) The SI unit of time is:

- Hour
- Minute
- ✓ Second
- Day

(iv) Which of the following statement is not true?

- Force can make the things at rest to move.
- Force can stop the moving objects.
- Force can change the speed of a moving object.

✓ Force cannot change the direction of a moving object.

(v) The force between two oppositely charged object:

Gravity

Frictional force

✓ Electrostatic force

Magnetic force

(vi) A motor car covers 1000 m in 10 s with uniform speed. It's speed will be:

50 ms-1

✓ 100 ms-1

500 ms-1

1000 ms-1

(vii) Which one of the following statements is correct?

1 hr = 60 m

✓ 1 hr = 60 min

1 hr = 60 km

1 hr = 60 s

(viii) A car covers the same distance in the same intervals of time, its speed is:

✓ Uniform.

Variable

Average

Moderate

(ix) It is the measure of how fast or slow an object moves per unit time.

- ✓ Speed
- Displacment
- Distance
- Force

(x) A car continuously covers 5 m in 2 s, 10 m in 4 s, 15 m in 6 s, 20 m in 8 s, and so on. The distance-time graph will show:

- ✓ Uniform speed
- Variable speed
- Average speed
- High speed

Exercise (p.125): 8.2 Define the following.

1. Force

Force is a push or pull that we apply on an object. Push and pull make the things move or stop or change their direction; friction, gravity, tension and normal force are also examples of force. Force can make the objects at rest to move, stop the moving objects, change the speed of a moving object, change the shape of an object, and change the direction of a moving object.

2. Gravity

Gravity is the Earth's pull on the objects. The force of gravity pulls objects towards the centre of the Earth — for example, when a ball is left at any height, it moves towards the ground due to gravitational force, and detached leaves or fruits from a tree fall on

the ground because of gravity.

3. Speed

Speed is a measure of how fast or slow an object is moving. It is calculated as the distance travelled by an object divided by the time taken to travel that distance: if speed is denoted by 'v', distance by 'S' and time by 't', then Speed (v) = distance (S) / time (t). For example, if a person runs a distance of 100 metres in 20 seconds, their speed is 100 m / 20 s = 5 metres per second.

4. Newton

Newton (N) is the SI unit of force, named after Sir Isaac Newton. One Newton (1 N) is defined as the force required to accelerate a mass of one kilogram (1 kg) by one metre per second square (1 m/s^2), written mathematically as 1 N = 1 kg × 1 m/s^2.

Exercise (p.125): 8.3 Write the SI units of the following.

1. Force

Force — Newton (N).

2. Distance

Distance — metre (m); it is also measured in units such as kilometres (km) and miles (mi).

3. Time

Time — second (s); it is also measured in units such as minutes (min) or hours (h).

4. Speed

Speed — metre per second (m/s or ms-1); the unit of speed is obtained by dividing the unit of distance, kilometre (km) or metre (m), by the unit of time, hours (hr) or second (s), giving kilometre per hour (km/hr) or metre per second (m/s).

Exercise (pp.125–126): 8.4 Constructed Response Questions.

1. Classify the following into two categories as contact and non-contact force: Friction, gravity, tension in rope, electrostatic force, magnetic force.

Contact force (forces when two objects are physically touching each other): friction (for example, when a box is made to slide across a table, friction acts between the box and the surface of the table) and tension in rope (the force transmitted through a rope, string or wire when pulled by forces acting from opposite sides). Non-contact force (forces that act on an object through a distance, i.e. without direct physical contact between the objects): gravity (the Earth's pull on objects), electrostatic force (when two charged objects are brought close, they repel or attract each other without touching each other) and magnetic force (magnets exert force on other magnets or magnetic materials without touching them).

2. Briefly describe the effect of non-contact force on each of the following: (a) Falling of an apple from the tree (b) A football that is kicked (c) Iron pins are attracted towards a magnet.

(a) Falling of an apple from the tree: this is the effect of the non-contact gravitational force — the force of gravity pulls the apple towards the centre of the Earth, so the detached apple falls

on the ground without anything touching it. (b) A football that is kicked: the kick is a push applied by contact — the force makes the football at rest begin to move (and a force applied on a moving football can stop it or change its speed or direction). (c) Iron pins are attracted towards a magnet: this is the effect of the non-contact magnetic force — a magnet exerts force on magnetic materials without touching them, so the iron pins are attracted towards the magnet.

3. The detail of a 12 hours journey of a car with a covered distance as shown in the table: 0–2 hours: 90 km; 2–6 hours: 150 km; 6–12 hours: 240 km. Determine the average speed of the car.
Step 1 — Total distance covered = 90 km + 150 km + 240 km = 480 km. Step 2 — Total time taken = 12 hours. Step 3 — Average speed = Total distance covered / Total time taken = 480 km / 12 hours = 40 km/h. So the average speed of the car is 40 kilometres per hour.

Exercise (p.126): 8.5 Investigate: toy train running down a slope. (a) Use the data in the Table to plot a graph of distance on Y-axis against time on X-axis. Time in seconds: 0, 2, 4, 6, 8, 10. Distance of train in metres: 0, 10, 20, 30, 40, 50.
Plot time on the X-axis (horizontal axis) and distance on the Y-axis (vertical axis) and mark the points from the table: (0 s, 0 m), (2 s, 10 m), (4 s, 20 m), (6 s, 30 m), (8 s, 40 m), (10 s, 50 m). Joining the points gives a straight line passing through the origin (0, 0) with a constant positive slope.

(b) State what the graph tells you about the motion of the toy train: (i) Starting Point (ii) From 2 to 4 seconds (iii) From 8 to 10 seconds.
The graph is a straight line with a constant positive gradient, which indicates that the object is moving at a constant (uniform) speed, since the distance covered increases linearly with time. (i) Starting point: at time 0 seconds the distance is 0 metres — the train starts from the origin (starting point), and the line passes through (0, 0). (ii) From 2 to 4 seconds: the distance increases from 10 m to 20 m, i.e. the train covers 10 m in 2 s at a speed of 10 m / 2 s = 5 m/s, and the straight line continues with the same slope, so the train is still moving at uniform speed. (iii) From 8 to 10 seconds: the distance increases from 40 m to 50 m, again 10 m in 2 s at 5 m/s — the line remains straight with the same slope, so the train keeps moving at the same constant speed throughout the whole run (overall speed = 50 m / 10 s = 5 m/s).

Note: In exercise 8.4 question 2 the book's header says 'the effect of non-contact force', but item (b) 'A football that is kicked' is a contact-force example (a kick is a push applied by contact, which makes the football at rest move); it is answered here by its actual force, and the mismatch is stated rather than silently corrected. The empty two-column table printed under question 2 on p.125 belongs to question 1 (the contact / non-contact classification). In MCQ (ix) the book prints 'Displacment' for 'Displacement'; the standard spelling is used in the options here.

Unit 9: Waves and Energy

Exercise (p.144): Multiple Choice Questions — 9.1 Encircle the correct option.

(i) The back and forth motion of the objects is called:

- Speed
- ✓ Vibrations
- Moving around
- Circular motion

(ii) Which energy of the vibrating object is transferred to the surrounding medium?

- Potential energy
- ✓ kinetic energy.
- heat energy.
- Mechanical energy.

(iii) What type of wave does not need medium to travel through?

- Mechanical
- ✓ Electromagnetic
- Sound
- Transverse

(iv) In which of the mediums, speed of sound is 343 metre per second?

✓ Air

Water

Iron

Wood

(v) Which sequence of visible colour is correct from longest wavelength to short wavelength?

Violet, indigo, blue, green, orange, yellow, red

✓ Red, orange, yellow, green, blue, indigo, violet

Red, yellow, green, orange, violet, blue, indigo

Red, yellow, green, orange, violet, indigo, blue

Exercise (p.144): 9.2 Match the statements of column A with the relevant statements of column B.

Glass →

5640 metre per second (the unit's table in section 9.4.2: Glass — 5640 metres per second).

Iron →

5130 metre per second (the unit's table in section 9.4.2: Iron — 5130 metres per second).

Air →

343 metre per second (the unit's table in section 9.4.2: Air — 343 metres per second).

Water →

1482 metre per second (the unit's table in section 9.4.2: Water —

1482 metres per second).

Exercise (p.145): 9.4 Write short answers.

(i) What is sound?

Sound is a form of energy. It is produced by vibrating objects, and it travels in the form of waves. Sound energy is in fact the kinetic energy of vibrating bodies: high sounds have more energy and low sounds have less energy.

(ii) What is meant by time period of a wave?

Time period is the time taken for one complete wave to pass a fixed point. It is represented by T, and it measures the duration it takes for a wave to complete one full oscillation. Time period and frequency are inversely proportional to each other: T = 1/f and f = 1/T (for example, an object producing a sound with a frequency of 50 Hz has a time period of 1/50 = 0.02 second).

(iii) Define the term echo?

An echo is the reflection of sound waves off a surface back to the listener's ears. Echoes occur when sound waves encounter a surface and reflect off it: when shouting in a large empty room or against a cliff, one can hear an echo, because the sound waves produced by the shouter travel to the nearest surface, reflect back, and reach the listener's ears, creating the sensation of hearing the sound again. An echo is heard clearly when the reflecting surface is 17 metres away and the reflecting surface is hard; a soft surface, such as a cushion, absorbs the sound waves and does not bounce them back.

(iv) What is needed by the sound to travel?

A material medium is needed by the sound to travel — it may be in any of the solid, liquid or gaseous states. Sound cannot travel in vacuum: a number of explosions are continuously taking place on the surface of the Sun, but we do not hear them on the Earth because there is a vacuum (no material medium) between the Sun and the Earth.

(v) State the wavelength range of visible light.

The wavelengths ranging from 700 nm to 400 nm in the electromagnetic spectrum make visible light which is detected by our eyes — that is, the wavelength range of visible light is 400 nm to 700 nm. When a beam of visible light is made to pass through a glass prism, it splits up into seven colours: violet, indigo, blue, green, yellow, orange and red.

Exercise (p.145): 9.5 Differentiate between the following.

(i) Mechanical wave and electromagnetic wave

Mechanical waves require a medium (solid, liquid or gas) to propagate: they transfer energy through the medium by causing particles of the medium to oscillate from their equilibrium position — examples include sound waves travelling through air and water waves travelling through water. Electromagnetic waves do not require a medium and can propagate through a vacuum: they are produced by the vibrations between an electric field and a magnetic field, and they consist of varying electric and magnetic fields that oscillate perpendicular to each other as they travel —

the light wave is an example of an electromagnetic wave.

(ii) Transverse wave and longitudinal wave

In a transverse wave, the particles of the medium vibrate (oscillate) perpendicular to the direction of wave propagation; the highest point of such a wave is called the crest and the lowest point is called the trough, and a complete wave cycle has both a crest and a trough — examples are waves on the surface of water and on a moving rope. In a longitudinal wave, the particles of the medium vibrate (oscillate) parallel to the direction of wave propagation — the particles move back and forth along the same line as the wave travels, and these waves exhibit compressions and rarefactions as they propagate through the medium — sound waves are a common example of longitudinal waves.

(iii) Wavelength and frequency

Wavelength is the distance between two successive crests or troughs of a wave; it is measured in metre (m) and is represented by the symbol λ (pronounced as Lambda). Frequency is the number of complete waves passing a fixed point in one second (also defined as the number of waves produced in one second); it is represented by the letter 'f' and measured in hertz (Hz). Frequency and time period are inversely related (f = 1/T).

(iv) Pitch and loudness

Pitch is the characteristic by which a shrill sound can be distinguished from a grave sound: it is how shrill or grave a sound feels to the human ear, and it is associated with the frequency of

the sound wave — a high-frequency sound wave corresponds to a high-pitched sound like a ringing bell, whereas a low-frequency sound wave corresponds to a low-pitched sound like a beaten drum (voices of children are of high pitch as compared to elders, voices of men are of low pitch as compared to women, and voices of birds are of high pitch than animals). Loudness is the characteristic of sounds due to which we distinguish between soft or faint and loud sounds: it is determined by the energy present in given sound waves and depends on the amplitude of the sound waves (loudness increases with amplitude and is directly proportional to the square of the amplitude — if the amplitude of a wave is doubled, its loudness becomes four times), the area of the sound source (a vibrating drum with large surface area produces a loud sound as compared with a vibrating drum with small surface area), the frequency of the sound waves, the distance from the sound source (loudness becomes fainter as we move away from the source), and the sensitivity of the ear.

(v) Sound waves and light rays

Sound waves are mechanical, longitudinal waves: they require a material medium (solid, liquid or gas) to travel, they travel by compressions and rarefactions of the medium's particles, and their speed in air is 343 metres per second. Light rays are electromagnetic, transverse waves: they do not require a medium and can propagate through a vacuum, they consist of varying electric and magnetic fields oscillating perpendicular to each other, and light travels much faster than sound — which is why we observe the flash of lightning earlier than its sound.

Exercise (p.145): 9.6 Constructed Response Questions.

(i) Sound is a form of energy. Explain.

Sound is a form of energy because it is produced by vibrating objects and it transfers energy from one place to another: a wave is any disturbance that carries energy through matter or vacuum (empty space), and waves transfer energy from one place to another without transferring the material (medium) through which they travel. Sound energy is in fact the kinetic energy of vibrating bodies — high sounds have more energy and low sounds have less energy. When we clap, it makes the air particles vibrate to produce waves, and these waves spread in all directions and reach our ears, which translate these waves in such a form that our brain feels them as sound. The energy transfer can be seen directly: the vibrating prongs of a tuning fork transfer their kinetic energy to air molecules, and the air molecules transfer this energy to water molecules, which begin to move and produce ripples or waves on the water surface.

(ii) How does sound produce?

Sound is produced by vibrating objects. When we speak, the vibration of our tongue creates waves in the air. When we beat a drum, sound waves are produced due to vibrations of the drum's diaphragm. When we clap, it makes the air particles vibrate to produce waves. When a tuning fork is struck against a rubber pad, its prongs vibrate, and bringing it close to the ear lets us feel the vibration and hear the sound. In short, any vibrating object — a drum's diaphragm, a tuning fork's prongs, the strings of a guitar,

our vocal organs — produces sound.

(iii) How does sound travel?

Sound needs a medium to travel: it travels in the air or other material mediums in the form of waves. Any vibrating object transfers its kinetic energy to the molecules of the surrounding air or other medium, which begin to vibrate also; these molecules collide with the neighbouring molecules, which too begin to vibrate, and in this way the sounds of vibrating objects cause waves in the surrounding medium which travel ahead in the medium. In air, the rapid vibrations of a drum's diaphragm cause disturbances that spread out in all directions: the upward movement of the diaphragm pushes the nearby air particles, creating a high-pressure zone called a compression, and when the diaphragm moves inward from its normal position, the air particles relax and spread out, creating a low-pressure zone called a rarefaction; as the diaphragm continues to vibrate, a series of alternate compressions and rarefactions move out as sound waves. In sound waves, the particles of air vibrate back and forth, while the sound waves travel in the forward direction (as shown by a candle flame flickering back and forth in front of a loudspeaker).

(iv) Speed of sound is different in different mediums. Why?

The speed of sound is different in different mediums because sound travels by the vibration of the medium's particles passed from one particle to the next, and the particles are packed differently in different mediums: in solids the particles are closely packed, so the vibrations are passed on quickly and sound travels fastest (glass

5640 metres per second, iron 5130 metres per second); in liquids the particles are less closely packed, so sound travels slower (water 1482 metres per second); and in gases the particles are far apart, so the vibrations are passed on most slowly (air 343 metres per second). The unit's table (section 9.4.2) gives: Air 343, Water 1482, Concrete 3200, Iron 5130 and Glass 5640 metres per second.

(v) Describe applications of electromagnetic waves.

Depending upon the wavelength or frequency range, the applications of electromagnetic waves in various fields of daily life are: Radio waves — used for communications such as radio, telephone and television. Microwaves — used for heating purposes in microwave ovens, e.g. for cooking food and heating water and food stuffs, and also for communication. Infra-red (IR) waves — used for short-range communication such as remote controls. Visible light — it illuminates things as we can see them through our eyes. Ultraviolet (UV) rays — used for sterilization, as they kill bacteria. X-rays — used to examine the body skeleton or hard tissues. Gamma rays — used to kill cancer cells during gamma-ray radiotherapy.

(vi) Why do we observe the flash of the thunder earlier than its sound?

We observe the flash of the thunder (lightning) earlier than its sound because sound travels slower than light. Thunder is the sound produced by the rapid expansion and contraction of air surrounding a lightning bolt. When lightning occurs, the light reaches our eyes almost instantaneously, but the sound takes a

longer time to reach our ears; the delay between seeing the lightning and hearing the thunder depends on the distance between the lightning bolt and the observer.

Note: MCQ (i) as printed labels its options 'a. Speed, b. Vibrations, b. Moving around, d. Circular motion' — the third option is mislabelled 'b.'; the options are rendered here in printed order with the intended a/b/c/d labels, and Vibrations (b) is ticked. The exercise numbering jumps from 9.2 (p.144) to 9.4 (p.145) — no section 9.3 is printed in the exercise; the section numbers are kept as printed.

Unit 10: Heat and Temperature

Exercise (pp.164–165): Multiple Choice Questions — 10.1 Encircle the correct option.

(i) Heat flows:

- From cold region to hot object
- ✓ From hot region to cold region
- From ice to liquid water
- Do not flow

(ii) Boiling point of water is:

- 0 °C
- 32 °F
- 100 °F
- ✓ 373 K

(iii) Centigrade scale is also called:

- Fahrenheit scale
- Kelvin scale
- Absolute scale
- ✓ Celsius scale

(iv) Melting point of ice is:

- 0°F
- ✓ 32°F
- 32°C

0 K

(v) Temperature in K = Temperature in °C + ...

32

100

212

✓ 273

(vi) 68 °C = ... °F

100

✓ 154

212

273

(viii) The process of heat transfer which does not require any medium is called:

Conduction

Convection

✓ Radiation

Insulation

(ix) The gaps are left after every few metres in the railway tracks, to avoid de-tracking in:

✓ Summer due to expansion

Winter due to expansion

Summer due to contraction

Winter due to contraction

(x) The rise of mercury level in the thermometer is due to ..... of mercury.

- Evaporation
- Contraction
- ✓ Expansion
- Sublimation

Exercise (p.165): 10.2 Give short answers.

1. Define plasma.

Plasma is the fourth state of matter, besides solid, liquid and gas. The unit names plasma as the fourth state of matter in section 10.1; in standard terms it is the state in which matter exists at extremely high temperatures, with its particles ionised (as in the Sun and stars).

2. Define temperature.

Temperature is the degree of hotness or coldness of an object. According to the Kinetic Molecular Theory, temperature is a measure of the average kinetic energy of the particles in a substance: a higher temperature indicates that the particles have higher average kinetic energy, meaning they move faster on average, and a lower temperature indicates that the particles have lower average kinetic energy. Temperature is measured in degrees Celsius (°C), Fahrenheit (°F) or Kelvin (K).

3. Name three modes of heat transfer.

The three modes of heat transfer are conduction, convection and radiation.

4. Give three examples of good conductors of heat.

Iron, copper and silver (the unit also lists aluminum): metals are good conductors of heat because, in metals, heat is not only conducted by vibrating atoms but also by the free electrons.

5. Give three examples of bad conductors of heat.

Wood, rubber and glass: non-metals are bad conductors of heat as they do not have free electrons in them (handles of cooking utensils are made from such materials).

Exercise (p.165): 10.3 Constructed Response Questions.

1. Gases expand fast as compared to solids and liquids. Explain Why?

Gases expand fastest because their particles move freely in all directions and are far apart with weak forces between them. When a gas is heated, its particles gain kinetic energy and move faster, and this increased movement causes the particles to spread out further apart, leading to an expansion of the gas that is very noticeable — gases can occupy much larger volume than solids or liquids. In solids, by contrast, the particles vibrate without leaving their position and are closely packed, so heating only increases their vibratory motion and pushes them slightly apart — the expansion of solids is usually very small and not easily noticeable. In liquids the particles move randomly without leaving the liquid surface, so the expansion of liquids on heating is more noticeable than solids but still less than gases. Hence the order of expansion on heating is: gases fastest, then liquids, then solids slowest.

2. Describe the expansion of three states of matter on heating, and contraction on cooling in term of particles.

On heating, the kinetic energy of the particles of a substance increases, so they move faster and push apart from each other, and the substance expands; on cooling, the kinetic energy of the particles decreases, so they move less and come closer together, and the substance contracts. Solids: when a solid is heated, the vibratory motion of its particles increases, which increases their kinetic energy, and this increase in vibration causes the particles to push apart from each other, leading to an expansion of the solid (usually very small); when a solid is cooled, the vibratory motion of its particles decreases and they lose kinetic energy, allowing the particles to move closer together, leading to a contraction of the solid (for example, a heated metal ball does not pass through a ring, but after cooling it passes through again). Liquids: when a liquid is heated, the motion of its particles increases and they gain kinetic energy, causing the particles to spread out further from each other, resulting in an expansion of the liquid (more noticeable than solids but less than gases); when a liquid is cooled, its particles lose kinetic energy and move less freely, allowing the particles to come close together, leading to a contraction of the liquid (for example, the level of coloured water in a flask tube rises on heating beyond its original position and returns to its original position on cooling). Gases: when a gas is heated, its particles gain kinetic energy and move faster, causing the particles to spread out further apart, leading to a very noticeable expansion; when a gas is cooled, its particles lose kinetic energy and move slower, allowing

the particles to come closer together, resulting in a contraction of the gas (for example, when the air inside a balloon is cooled, it contracts and occupies less space than when it was heated).

Exercise (p.165): 10.4 Write notes on the followings.

a. Kinetic Molecular Theory

Kinetic Molecular Theory: the particles forming all kinds of matter are always in a state of motion, and heat energy is associated with the random motion of particles in a substance. When heat is added to a substance, the kinetic energy of its particles increases, causing them to move faster and collide with each other more frequently; heat causes substances to expand, change state (e.g. melt, vaporize) and increase in temperature. Temperature is a measure of the average kinetic energy of the particles in a substance and is directly related to it: a higher temperature indicates higher average kinetic energy (particles move faster on average), and a lower temperature indicates lower average kinetic energy. The total kinetic energy of all the particles in a substance is called its thermal energy or heat.

b. Thermal expansion

Thermal expansion: when a substance is heated, the kinetic energy of its particles increases and they push apart from each other, so the substance expands. The expansion of solids is usually very small; the expansion of liquids is more noticeable than solids but less than gases; the expansion of gases is very noticeable. Daily-life effects: metal rods used in bridges and railways expand on heating and contract on cooling and change the structures of bridges and

railways — in hot weather railway lines sag or bend due to thermal expansion; small gaps are left after every few metres in concrete roads and paths because concrete expands in summer heat and would otherwise rupture; gaps are left at one end of bridges, supported by rollers, so the bridge can slide on expansion without cracking the concrete; overhead power and telephone wires are strung loosely from pole to pole so they do not sag too much in summer; a metal tyre is fixed over a wooden wheel by heating it so its diameter increases, and on cooling it contracts and fits tightly; a hot rivet inserted through steel plates contracts on cooling and forces the plates closer, firmly gripping them together.

c. Thermal contraction

Thermal contraction: when a substance is cooled, the kinetic energy of its particles decreases and they move closer together, so the substance contracts. Examples: a heated metal ball that cannot pass through a ring passes through it again after cooling; the level of coloured liquid in a flask tube falls back to its original position when the system cools to room temperature; the air inside a balloon contracts on cooling and occupies less space; overhead power and telephone wires contract in winter and become tight and can snap, so allowance is made for contraction when they are strung; a hot rivet contracts on cooling and pulls the steel plates tightly together; mercury or alcohol in a thermometer contracts and its level falls in the tube at a cold place.

d. Conduction

Conduction: transfer of heat from the hot end of a material object

to the cold end due to increased collision of particles. Particles within the part of a material which is heated gain heat energy and begin to vibrate more vigorously; these vibrating particles collide with neighbouring particles and transfer kinetic energy to them, and this process continues throughout the material, resulting in the transfer of heat from regions of higher temperature to regions of lower temperature. Metals are good thermal conductors because their atoms are closely packed with free electrons: the free electrons move freely throughout the material and transfer energy to neighbouring particles through collisions (electron conduction), so heat spreads quickly. Fluids (liquids and gases) are poor thermal conductors because their particles are much farther apart and there are no free electrons as in metals, so heat transfer in fluids is much slower and mainly driven by convection rather than conduction. Example: the tip of a metal spoon dipped in boiling water feels hot after 2–3 minutes, while the tip of a plastic spoon does not.

e. Convection

Convection: the transfer of thermal energy through the actual movement of molecules (liquids or gases) from one place to another. In heat convection, particles in a fluid gain heat and get their movement and kinetic energy increased; this makes them less dense and they move towards the cold region of the fluid, and such movement of fluid carries thermal energy from hotter regions to cooler regions. Boiling water in a pot is an example: when water at the bottom of the pot is heated, it becomes less dense and rises, and cooler water from the top of the pot then sinks to replace the

rising hot water; the same mechanism of convection takes place in geysers.

f. Radiation

Radiation: the transfer of thermal energy through electromagnetic waves. The radiation process does not require any material medium for heat transfer: particles with thermal energy emit electromagnetic waves (infrared radiation) due to their vibratory motion, these waves travel through space and can be absorbed by other objects to give them warmth. Heat from the Sun reaches the Earth through the radiation process — the Sun emits electromagnetic waves that travel through space and are absorbed by the Earth's surface. Similarly, the warmth felt from a fireplace or a hot stove is due to heat radiation emitted by the hot objects.

Exercise (pp.165–166): 10.5 Investigate.

1. Substances do not expand or contract at the same rate. Why?

Substances do not expand or contract at the same rate because their particles differ in arrangement, spacing and the strength of the forces between them, so the same rise or fall in temperature changes their motion by different amounts. The unit shows this across the states of matter: solids expand least (their particles only vibrate about fixed positions in a close-packed arrangement), liquids expand more (their particles move randomly), and gases expand most (their particles move freely in all directions) — and even among solids, different metals expand at different rates, which is why expansion gaps must be left in railway tracks, concrete roads and bridges, why a metal tyre is heated before

fitting it over a wheel, and why a hot rivet grips plates firmly as it cools and contracts.

2. Give the uses of expansion and contraction of liquids.

The main use of the expansion and contraction of liquids in the unit is the thermometer: thermometer bulbs contain mercury or alcohol, and these liquids expand and rise up in the tube when the bulb comes into contact with a hot object or the human body, and they contract and fall down in the tube on becoming cold — the expansion and contraction of the mercury or alcohol in the glass tube thus indicates the rise and fall of temperature. The same behaviour is shown in the unit's flask activity: on heating, the level of coloured water in the glass tube rises beyond its original position, and on cooling back to room temperature it returns to its original position.

3. An empty steel container is sealed and heated, which of the following properties of the gas is likely to increase?

- Mass
- ✓ Pressure
- Volume
- Density

Note: MCQ numbering in print jumps from (vi) to (viii) — no item (vii) is printed; the items are kept with their printed numbers, and the missing (vii) is stated here. MCQ (ii) deliberately omits 100 °C from the options (100 °F is given instead, which is wrong), so 373 K — the Kelvin boiling point of water — is ticked; likewise MCQ (iv)

omits 0 °C, so 32 °F — the Fahrenheit melting point of ice — is ticked. MCQ (vi): 68 °C = (68 × 9/5) + 32 = 154.4 °F, so 154 is ticked. In 10.2 the third item is misnumbered '2.' in print (after '2. Define temperature'); it is renumbered 3 here. In 10.5 item 3 (p.166), Pressure is ticked because the sealed steel container fixes the volume and the mass of the gas, so density (mass/volume) also stays fixed, while heating increases the kinetic energy and collisions of the gas particles and hence the pressure. 10.6 Project (p.166) is marked 'Teacher Guide' — the teacher facilitates students to investigate the strange/unusual behaviour of water during contraction and expansion. It is a classroom investigation with no single written answer in the book, so no model answer is given for it here.

Unit 11: Technology in Everyday Life

Exercise (pp.174–175): Multiple Choice Questions — 11.1 Encircle the correct option.

(i) Drip irrigation system reduces water wastage by minimizing:

- Absorption
- ✓ Evaporation
- Transpiration
- Transportation

(ii) Sprinkler irrigation system sprays water over:

- ✓ Leaves
- Roots
- Stem
- Flowers

(iii) Water conservation means:

- Wasting water
- Cleaning water
- ✓ Saving water
- Dropping water

(iv) The one which is not used in making pickles?

- Salt
- Pepper
- Vinegar

✓ Honey

(v) The essential one used in making mango pickle?

Mango slices

Salt

Vinegar

✓ All of these

(vi) A liquid material used to kill the germs is _____.

✓ Sanitizer

Honey

Vinegar

Pepper

(vii) Which one of the followings is not used in the preparation of apple jam?

Apple

✓ Salt

Sugar

Cinnamon

Exercise (p.175): 11.2 Write short answers.

1. What is the advantage of drip irrigation system?

The drip irrigation system delivers water directly to the roots of plants through small holes in the drip emitters, so it reduces water wastage by minimizing evaporation and runoff; it helps conserve water by ensuring proper hydration for plants, it is suitable for areas with limited water resources, and it increases agricultural

yields.

2. What type of irrigation is used in the areas with limited water resources?

The drip irrigation system is used in areas with limited water resources.

3. List the ingredients required for preparation of mango pickle.

Un-ripened mangoes (4–5 pieces), salt, vinegar (preservative), and a pickle jar with lid.

4. List the materials needed for making onion pickle.

Onion, 2 cinnamon sticks, 6 whole cloves, a pinch of crushed red pepper flakes, 1 cup (240 mL) of vinegar, 1/2 cup (120 mL) of freshly-squeezed lime juice, 1/2 cup (100 g) of sugar, and 1 tablespoon of common salt.

5. List the ingredients required for preparation of apple jam.

Sliced apple (2 bowls), sugar (1/3 cup), cinnamon (1 small stick), orange zest (1 tablespoon), lemon juice (preservative), and a bottle (jar).

Exercise (p.175): 11.3 Answer the following questions.

1. Explain drip irrigation.

Drip irrigation is the application of water only to the root zone of the plants. The system delivers water directly to the roots of plants through small holes in the drip emitters. It reduces water wastage by minimizing evaporation and runoff, it is suitable for areas with

limited water resources, it helps conserve water by ensuring proper hydration for plants, and it increases agricultural yields. A simple drip irrigation system can be made by hanging a plastic bottle (bottom end cut off) above a potted plant, fitting a drip pipe with a needle and valve/stopper so the needle drips water to the roots, passing the free end of the pipe through a hole in the bottle lid, sealing leakage with a glue gun, filling the bottle with water, and opening the valve so water drips to the roots; dripping can be stopped with the stoppers or valves.

2. Describe sprinkler irrigation.

Sprinkler irrigation is the application of water on the surface of crop and soil. The system distributes water over a larger area in a spray-like pattern: it covers more ground but leads to more water loss through evaporation. It is effective for larger agricultural fields, but it is not as water-efficient as drip irrigation systems, especially in regions with limited water availability. A simple sprinkler system can be made by placing a water tub or tank at a height with a hole slightly above its bottom, fixing a plastic pipe from the hole to the mouth of a plastic bottle, making small holes in the bottom of the bottle, hanging the bottle over the plants, and filling the tub with water (with dissolved chemical fertilizers) so the water flows through the pipe and sprinkles over the plants through the holes in the bottle.

3. Write the procedure for making mango pickle?

Materials: un-ripened mangoes (4–5 pieces), a pickle jar with lid, salt, vinegar (preservative). Procedure: (1) Take un-ripened

mangoes (4–5 pieces). (2) Wash them and cut them into pieces. (3) Place these mango pieces in the sun and let them dry. (4) Keep these dried mango pieces dipped in salt water overnight. (5) Pour these mango pieces into clean and sterilized jars containing vinegar solution (preservative). (6) Air-tight the jars with lids. (7) Mango pickles will be ready for use in 15 days.

4. Describe the procedure for making hand sanitizer.

A sanitizer is a liquid mixture used to kill germs. Materials: distilled water, glycerine, isopropyl alcohol, hydrogen peroxide, and a plastic or glass container. Procedure: (1) Take a plastic or glass container and add in it: (i) 4 small cups of isopropyl alcohol, (ii) 1/4 small cup of hydrogen peroxide, (iii) one small cup of distilled water, (iv) one teaspoonful of glycerine, (v) shake the container well and mix all the ingredients to make a uniform solution. (2) Hand sanitizer is ready for use.

5. Describe the procedure for making simple stethoscope.

A stethoscope is a medical device used for listening to the heart beat and counting the pulse rate of the human body. Materials: a Y-shaped tube, fine hollow plastic tubes, cotton, a balloon, scissors, insulation tape, and a funnel. Procedure: (1) Take a Y-shaped tube. (2) Cut two small pieces of fine hollow plastic tube. (3) Fix one end of a small fine hollow tube over one upper limb of the Y-shaped tube and one end of the second small hollow tube over the other upper limb of the Y-shaped tube. (4) Take two cotton swabs and fix them on the free ends of the two small fine hollow tubes fitted with the limbs of the Y-shaped tube. (5) Take another slightly long fine

hollow tube and fix its one end over the third limb of the Y-shaped tube and its second end over the stem of the funnel. (6) Take a balloon, cut its neck and wrap its lower part over the funnel. (7) Wrap insulation tape over the tubes and the funnel. (8) The instrument so made is a simple homemade stethoscope. (9) Use it for listening to the heart beat and counting the pulse rate of the human body.

Exercise (p.175): 11.4 Constructed Response Questions.

1. Describe the use of technology in agriculture.

Technology in agriculture is used above all in irrigation: drip and sprinkler irrigation systems are very useful technology for water conservation and increasing agricultural yields. The drip irrigation system delivers water directly to the roots of plants through small holes in the drip emitters, reducing water wastage by minimizing evaporation and runoff, and it is suitable for areas with limited water resources, helping conserve water by ensuring proper hydration for plants. The sprinkler irrigation system distributes water over a larger area in a spray-like pattern and is effective for larger agricultural fields, though it is not as water-efficient as drip irrigation, especially in regions with limited water availability. Understanding and demonstrating drip and sprinkler irrigation systems helps in raising awareness about water conservation methods in agriculture, and students design a model to demonstrate drip and sprinkler irrigation systems for conservation of water.

2. Describe the use of technology in food preservation.

Food preservation depends on modern technology. The flavour and nutritional value of food items like juices and jams, etc., is preserved by heating at a specific temperature for a certain duration to kill harmful microorganisms — this process is called pasteurization — and the heated juices or jams are then quickly cooled and packaged in sterilized containers to prevent contamination. Examples from the unit: orange juice is made by rinsing and peeling oranges, slicing them into four parts, grinding the slices in a juicer with 2–3 spoonfuls of sugar, heating the juice to around 85 °C for 5–10 seconds, cooling it and rapidly pouring it into clean and sterilized containers with air-tight seals to prevent bacterial contamination; pasteurized orange juice can be stored in the refrigerator for several weeks or further preserved through freezing. Apple jam is made by washing apples, removing the skin and cutting them into slices, mixing the sliced apple with sugar, a cinnamon stick, orange zest and lemon juice (preservative), cooking the mixture at medium heat, stirring from time to time until it comes to simmer, reducing the heat to medium-low and simmering for 20 to 30 minutes, and storing the jam in a jar. Mango pickles are made by washing un-ripened mangoes, cutting them into pieces, drying them in the sun, dipping them in salt water overnight, and pouring them into clean and sterilized jars containing vinegar solution (preservative), air-tight with lids, ready for use in 15 days. Onion pickles are made by dipping sliced onions in boiled water for 12 minutes, draining and transferring them into a clean and sterilized glass jar, heating cinnamon, cloves and red pepper flakes for 3 to 5 minutes, adding vinegar, lime juice, sugar and salt, heating until the sugar and salt are dissolved, pouring this

solution over the onions in the jar and closing it with the lid; the pickle is ready for use in about 20 days.

Exercise (p.176): 11.5 Investigate.

1. Prepare a simple stethoscope, use it for listening heartbeat and counting pulse rate of your five classmates and record your observations.

Prepare a simple stethoscope by the procedure in question 11.3 (5) above (Y-shaped tube, fine hollow plastic tubes, cotton swabs, funnel and balloon). Use it for listening to the heart beat and counting the pulse rate of your five classmates, and record your observations in the unit's table (serial number, name of classmate, heartbeat, pulse rate). The table is filled with the students' own observations during the activity, so no classmate data is given here.

Note: MCQ (vi) as printed leaves the blank as boxes ('is □□□□.'); it is rendered here as 'is _____.' 11.6 Project (p.176): prepare a model for drip irrigation — a classroom project with no single written answer in the book, so no model answer is given for it here.

Unit 12: Earth and Space

Exercise (p.188): Multiple Choice Questions — 12.1 Encircle the correct option.

(i) The force of attraction between any two masses:

- ✓ Gravity
- Friction
- Pull
- Push

(ii) What counteracts the gravitational pull and keeps planets revolving along a certain path around the Sun?

- Mass of planets
- Weight of planets
- ✓ Velocity of planets
- Volume of planets

(iii) Weight is measured in:

- kg
- km
- ✓ N
- m/s

(iv) Weightlessness is a situation where a mass has zero:

- Mass

✓ Weight

Velocity

Volume

(v) Seasons on the Earth's surface are caused by:

Spin motion of the Earth about its own axis

Orbital motion of the Earth around the Sun

Tilt of Earth on its axis

✓ The tilt of Earth and orbital motion of the Earth around the Sun.

(vi) What is the season on 22 June in the southern hemisphere?

Spring

Autumn

Summer

✓ Winter

(vii) The weight of an object on Earth's surface is 300 N. What will be its weight on the Moon's surface?

200 N

180 N

100 N

✓ 50 N

Exercise (p.189): 12.2 Answer the following questions briefly.

1. Explain gravitational force.

According to Isaac Newton, every object attracts every other object; such a force is called gravitational force or gravity. The larger the mass of an object, the stronger the gravitational force it

exerts, and the closer the two objects are, the stronger the gravitational force between them. The Earth pulls every object towards its centre, and on the other hand the object pulled towards the Earth's centre also pulls the Earth towards its own centre; the net result is a force of attraction between the Earth and the object. The greater the mass of an object, the greater is its pull on the other object, and as the Earth's mass is greater than the object placed on it, its pull on the object is also greater — that is why things dropped from some height move towards the Earth's centre.

2. How gravity keeps planets in the orbits?

The gravitational forces between massive objects such as the Sun and the planets, and the Earth and the Moon, keep the smaller objects in their orbits. Objects with smaller mass revolve with different velocities around objects with greater mass, and it is their velocity that counteracts the gravitational pull and keeps them revolving along a certain path. The path followed by an object while revolving around another object in space due to the force of gravity is called an orbit: it is the force of gravity that keeps the planets in their orbits around the Sun, and similarly moons revolve around the planets — the Moon orbits the Earth due to the force of gravity between them, which keeps the Moon in its elliptical orbit around the Earth. The unit's string-and-toy-car activity shows the same idea: the car tied by a string to a nail moves in a circle around the nail, and when the string is cut, the car flies off — the string, like gravity, holds the moving object in its circular path.

3. Differentiate between mass and weight.

Mass is a measure of how much 'matter' is present in an object; it is measured in kilograms (kg) or grams (g), and it remains constant — the value of mass does not change with gravitational force, whether it is measured at a point far away from the centre of the Earth or very close to its centre, and the mass of an object cannot be zero. Weight is the force exerted on an object due to gravity — the gravitational force acting on an object's mass; it is measured in newton (N), it depends on the value of gravitational field strength at the location of the object and varies with changes in gravitational force (it is different at different distances from the centre of the Earth), and the weight of an object may be zero. Mathematically, W = mg, where W is the weight of the object, m is its mass and g is the acceleration due to gravity: if acceleration due to gravity is constant, weight increases with the increase in mass, and if mass is constant, weight increases with the increase in gravity (for example, the weight of an object of mass 1 kg on the Earth's surface is about 10 N).

4. How are tides caused in seas?

Tides are the rising and falling of sea levels. They are caused by the gravitational pull of the Moon: the gravitational pull is stronger on the side of the Earth facing the Moon and weaker on the opposite side, and this difference in gravitational force creates a tidal force, causing the oceans to bulge towards and away from the Moon — producing high tides on both sides and low tides in between. These daily changes in the level of ocean water are also caused by the gravitational force of the Moon and the Sun as well as by the rotation of the Earth about its axis. Low tides help the fishermen

move forwards in the sea and high tides help them come back more easily. Tides used as a renewable energy source to generate electricity are called tidal energy.

5. How do seasons change?

Seasons change because of the Earth's annual revolution around the Sun together with the tilt of its axis. The Earth completes one revolution around the Sun in 365 days, which we call one year, and the path followed by the Earth during its revolution is called its orbit. The Earth's axis is tilted at an angle of approximately 23.5 degrees relative to its orbit around the Sun, and as the Earth orbits the Sun, this tilt causes different parts of the Earth to receive varying amounts of sunlight throughout the year, leading to the changing of seasons. The summer solstice occurs around June 22nd in the Northern Hemisphere, when the North Pole is tilted towards the Sun, resulting in the longest day of the year and the start of summer; conversely, the winter solstice occurs around December 22nd in the Northern Hemisphere, when the North Pole is tilted away from the Sun, resulting in the shortest day of the year and the start of winter. The spring equinox occurs around March 21st, when the tilt of the Earth's axis is such that neither pole is tilted towards or away from the Sun, resulting in nearly equal lengths of day and night; the autumn equinox occurs around September 23rd, marking the start of fall, with a similar balance between day and night.

6. What will be the season in Australia when there is summer in Pakistan?

Winter. Pakistan is in the Northern Hemisphere and Australia is in

the Southern Hemisphere, and the seasons in the two hemispheres are opposite: when the North Pole is tilted towards the Sun (around June 22nd), the Northern Hemisphere has summer while the Southern Hemisphere has winter. So when there is summer in Pakistan, there is winter in Australia — as confirmed by MCQ (vi): on 22 June the season in the southern hemisphere is winter.

Exercise (p.189): 12.3 Constructed response questions.

1. What are constellations? How are they helpful in mapping the sky?

A constellation is a specific pattern or shape formed by a group of stars in the sky: we see a huge number of stars in the sky at night, stars look as groups, and some groups look like a specific pattern or shape. Constellations are helpful in mapping the sky because it is easy to recognize a star by its constellation — a constellation is an area of the sky which can be used for reference purposes for the whole sky. There are 88 constellations covering the whole sky, and in old days they were used to set direction for navigation in the sea. Some famous constellations are Ursa Major (or big dipper), Great Bear, Leo and Orion: Ursa Major or big dipper consists of seven stars — four stars make the bowl and three stars form its handle, and a bright star on the end of the bowl points to the pole star; Leo is seen in the sky from March to May each year. The constellations that appear during different parts of the year are caused by the Earth's revolution around the Sun: as the Earth orbits the Sun, its position changes relative to the background stars, causing different constellations to be visible at different times of the year — for example, during the summer months in the Northern Hemisphere,

constellations such as Cygnus and Aquila are more prominent, while during the winter months, constellations such as Orion and Taurus dominate the night sky.

2. The Earth and the Moon have gravitational force (force of attraction) between them. Why does not Moon fall on the Earth?
The Moon does not fall on the Earth because its orbital velocity counteracts the Earth's gravitational pull. Objects with smaller mass revolve with different velocities around objects with greater mass, and it is their velocity that counteracts the gravitational pull and keeps them revolving along a certain path (orbit): the Moon orbits the Earth due to the force of gravity between them, and this gravitational force keeps the Moon in its elliptical orbit around the Earth. The unit's string-and-toy-car activity shows the same idea: a toy car tied by a string to a nail moves in a circle around the nail, and when the string is cut, the car flies off its path — so as long as the Moon keeps its orbital velocity, gravity holds it in its orbit instead of pulling it down to the Earth.

3. Which one has a greater pull on oceans water. Sun or Moon? Give reasons.
The Moon has a greater pull on the oceans' water than the Sun, because the Moon is closer to the Earth than the Sun. Although the Sun is far more massive, gravitational force also depends on distance: the closer the two objects are, the stronger the gravitational force between them — so the Moon pulls the ocean water more strongly than the Sun does. The unit states this directly: 'The Moon is closer to the Earth than the Sun. It pulls the ocean

water more strongly, as compared to the Sun.' Tides are therefore caused by the gravitational forces exerted by the Moon and, to a lesser extent, the Sun on Earth's oceans. When the Sun and Moon are lined up with the Earth, their combined gravitational pull creates higher tides called spring tides — this happens when the Moon is in its new phase or full phase.

4. What happens to mass and weight of an astronaut when he leaves the Earth and travels to the Moon? Explain.

His mass does not change, but his weight decreases. The mass of an object remains the same whether it is on Earth, the Moon, or any other celestial body, because mass is the amount of matter in the object and does not change with gravitational force. His weight, however, depends on the gravitational field strength: as he moves far from the Earth, his weight decreases, and as he leaves the Earth's gravitational field, his weight becomes zero — the situation he feels is called weightlessness, and things float around inside a satellite and spaceships which orbit the Earth as they have no weight. When he lands on the Moon's surface, his weight will be according to the gravitational force between the Moon and himself: the gravitational pull on the Moon is much weaker than on Earth, about one-sixth (1/6) as strong, so objects on the surface of the Moon weigh much less than they do on Earth — for example, a person who weighs 60 N on Earth would weigh only 10 N on the Moon, and an object weighing 300 N on Earth weighs 50 N on the Moon (MCQ vii). Astronauts on the Moon experience a sensation of weightlessness or 'free-fall' because the gravitational force pulling them towards the Moon's surface is significantly reduced.

Exercise (p.189): 12.4 Investigation: use library and internet facility and make the following studies.

1. Do other stars have planets like the Sun?

Yes — other stars have planets like the Sun. This question is set by the book as a library and internet investigation (it is not answered in Unit 12's text): astronomers have discovered thousands of planets orbiting other stars, called exoplanets, using telescopes and space missions, so planetary systems are not unique to our Sun.

2. What are galaxies? How does a constellation differ from a galaxy?

A galaxy is a huge system of billions of stars together with gas and dust, all held together by gravity — for example, our own Milky Way galaxy, which contains our Sun and its planets. A constellation differs from a galaxy in that a constellation is only a pattern of stars as seen from the Earth: it is a specific pattern or shape formed by a group of stars in a particular area of the sky, used for reference and mapping the sky, and its stars are not physically bound together — they only appear grouped from our viewpoint and may lie at very different distances from Earth. A galaxy, by contrast, is a real, physically bound system of stars. (This study is set by the book as a library and internet investigation; the constellation part follows Unit 12 section 12.6, while the galaxy part comes from the directed library/internet study.)

Note: MCQ (vii): the Moon's gravitational pull is about one-sixth

(1/6) of the Earth's, so 300 N ÷ 6 = 50 N on the Moon. Glossary (pp.190–196) is a reference glossary of terms with definitions and carries no exercises, and pp.197–199 are the Index; both are noted as not reprinted, as with the فرہنگ handling in the Urdu books. Author details on p.196 are not exercises. 12.5 Project (p.189): manage a seminar in your school or classroom and conduct lectures of experts in relevant fields on (a) Weightlessness, (b) Constellations, (c) The Earth's Gravity and the Moon's Gravity — a classroom project with no single written answer in the book, so no model answer is given for it here.